

Dévissage for Algebraic K-theory of Small Stable ∞ -categories

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ABSTRACT

In this article, we extend the theorem of heart[Bar15], which implies Quillen's dévissage theorem by [Efi25], to generic small stable ∞ -categories. To be precise, we establish some sufficient conditions under when an exact functor between stable ∞ -categories induces isomorphisms of non-negative K -groups.

Keywords: Category Theory K-Theory

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INTRODUCTION

Algebraic K -theory provides fundamental tools to investigate the structure of stable ∞ -categories, revealing deep connections between categorical decompositions and invariants arising from homotopy-theoretic contexts. In this article, we establish new results characterizing algebraic K -groups under conditions generalizing classical dévissage arguments.

Our main contributions focus on exact functors between (idempotent-complete) stable ∞ -categories that satisfy suitable dévissage conditions. Building upon this framework, we prove that an exact functor induces isomorphisms on all higher K -groups K_n for $n \geq 1$ whenever it satisfies a weak dévissage condition and is fillable. Furthermore, if the stronger dévissage condition is met, the isomorphism extends to K_0 . In addition, we establish vanishing results for higher K -groups of fillable stable ∞ -categories themselves, uncovering new structural constraints.

Theorem A. (Theorem 4.12) Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be a n -fillable exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ satisfies the weak dévissage condition (resp. dévissage condition), it induces isomorphisms $K_i(\mathcal{A}) \xrightarrow{\cong} K_i(\mathcal{C})$ for all $1 \leq i \leq n-1$, an epimorphism $K_n(\mathcal{A}) \twoheadrightarrow K_n(\mathcal{C})$ and a monomorphism (resp. an isomorphism) $K_0(\mathcal{A}) \hookrightarrow K_0(\mathcal{C})$.

Theorem B. (Theorem 4.15) Let $\mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be a small n -fillable idempotent-complete ∞ -category. Then $K_i(\mathcal{C}) = 0$ for all $1 \leq i \leq n$.

Theorem C. (Theorem 4.30) Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be a strongly 1-fillable exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ satisfies the dévissage condition, it induces isomorphisms $K_n(\mathcal{A}) \xrightarrow{\cong} K_n(\mathcal{C})$ for all $n \geq 0$.

Overview of this article:

- In Section 1, we introduce basic definitions and properties of stable ∞ -categories.
- In Section 2, we formalize the (weak) Dévissage condition for an exact functor. It concerns whether the objects in the image are enough to generate the whole ∞ -category.
- In Section 3, we develop the categorification of fibers and loops for universal localizing invariant. It is used to describe the higher K -groups of our target by K_0 -groups of some other ∞ -categories.
- In Section 4, we discuss the fillability of exact functors and present the main results concerning algebraic K -theory. This condition determines whether the class of cofiber sequences in the image are enough to describe the class of cofiber sequences in the whole ∞ -category.
- In Section 5, we show how to use our results to prove the theorem of heart.

1 PRELIMINARIES ON STABLE ∞ -CATEGORY AND EXACT FUNCTORS

1.1 Stable ∞ -category

Beyond ordinary categories, the landscape of category theory expands with the introduction of higher categories, including the concept of ∞ -categories, which play an important role in K -theory. Bergner [Ber06] wrote a survey of different models of ∞ -categories. The most widely used model, due to Joyal [And02] and Lurie [Lur09], defines ∞ -categories as *weak Kan complexes* (or *quasi-categories*).

Definition 1.1. [Lur17, Definition 1.1.1.4]

A *triangle* in a pointed ∞ -category $\mathcal{C}$ is a functor $\Delta^1 \times \Delta^1 \rightarrow \mathcal{C}$

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & & \downarrow g \\ 0 & \longrightarrow & Z \end{array}$$

A *fiber sequence* in $\mathcal{C}$ is a pullback triangle, and a *cofiber sequence* in $\mathcal{C}$ is a pushout triangle. We will abuse notation by writing $X \rightarrow Y \rightarrow Z$ for a triangle.

Definition 1.2. [Lur17, Definition 1.1.1.6]

Given a morphism $g : X \rightarrow Y$ in a pointed ∞ -category $\mathcal{C}$. A *fiber* of g is a fiber sequence

$$\begin{array}{ccc} W & \longrightarrow & X \\ \downarrow & & \downarrow g \\ 0 & \longrightarrow & Y \end{array}$$

Dually, a *cofiber* of g is a cofiber sequence

$$\begin{array}{ccc} X & \xrightarrow{g} & Y \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & Z \end{array}$$

We also refer to W and Z simply as the fiber and cofiber of g , respectively, and write $W = \text{fib}(g)$ and $Z = \text{cofib}(g)$.

Definition 1.3. [Lur17, Definition 1.1.1.9, Proposition 1.1.3.4]
A *stable* ∞ -category is a pointed ∞ -category satisfying:

- It is finite complete and finite cocomplete.
- A square is a pushout if and only if it is a pullback.

Or, equivalently:

- Every morphism admits both a fiber and a cofiber.
- A triangle is a fiber sequence if and only if it is a cofiber sequence.

Definition 1.4. Let $\mathcal{C}$ be an ∞ -category.

- (1) Two morphisms $f, g : X \rightarrow Y$ are *(homotopy) equivalent*, denote by $f \simeq g$, if $[f] = [g]$ in $\pi_0 \text{Map}(X, Y)$.
- (2) Two objects X, Y are *(homotopy) equivalent*, denote by $X \simeq Y$, if there are two morphisms $f : X \rightarrow Y, g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$. f, g are called *(homotopy) equivalences*.

We display some properties of stable ∞ -categories firstly.

Lemma 1.5. Suppose $X' \rightarrow X \rightarrow X'', Y' \rightarrow Y \rightarrow Y'',$ and $Z' \rightarrow Z \rightarrow Z''$ are three cofiber sequences in a stable ∞ -category $\mathcal{C}$.

- (1) Given a functor $f : \Delta^1 \times \Delta^1 \rightarrow \mathcal{C}$ represented by the diagram

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \end{array},$$

there is a unique morphism $X'' \rightarrow Y''$ extending f to a functor $\Delta^2 \times \Delta^1 \rightarrow \mathcal{C}$ of the form

$$\begin{array}{ccccc} X' & \longrightarrow & X & \longrightarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y' & \longrightarrow & Y & \longrightarrow & Y'' \end{array}.$$

- (2) Given a functor $f : \Delta^1 \times \Delta^1 \rightarrow \mathcal{C}$ represented by the diagram

$$\begin{array}{ccc} X & \longrightarrow & X'' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y'' \end{array},$$

there is a unique morphism $X' \rightarrow Y'$ extending f to a functor $\Delta^2 \times \Delta^1 \rightarrow \mathcal{C}$ of the form

$$\begin{array}{ccccc} X' & \longrightarrow & X & \longrightarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y' & \longrightarrow & Y & \longrightarrow & Y'' \end{array}.$$

- (3) Given a functor $f : \Delta^1 \times \Delta^2 \rightarrow \mathcal{C}$ represented by the diagram

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow & & \downarrow \\ Y' & \longrightarrow & Y \\ \downarrow & & \downarrow \\ Z' & \longrightarrow & Z \end{array}$$

where the two columns are cofiber sequences, there are unique morphisms $X'' \rightarrow Y''$ and $Y'' \rightarrow Z''$ extending f to a functor $\Delta^2 \times \Delta^2 \rightarrow \mathcal{C}$ of the form

$$\begin{array}{ccccc} X' & \longrightarrow & X & \longrightarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y' & \longrightarrow & Y & \longrightarrow & Y'' \\ \downarrow & & \downarrow & & \downarrow \\ Z' & \longrightarrow & Z & \longrightarrow & Z'' \end{array}$$

such that $X'' \rightarrow Y'' \rightarrow Z''$ is a cofiber sequence.

(4) Given a functor $f : \Delta^1 \times \Delta^2 \rightarrow \mathcal{C}$ represented by the diagram

$$\begin{array}{ccc} X & \longrightarrow & X'' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y'' \\ \downarrow & & \downarrow \\ Z & \longrightarrow & Z'' \end{array}$$

where the two columns are cofiber sequences, there are unique morphisms $X' \rightarrow Y'$ and $Y' \rightarrow Z'$ extending f to a functor $\Delta^2 \times \Delta^2 \rightarrow \mathcal{C}$ of the form

$$\begin{array}{ccccc} X' & \longrightarrow & X & \longrightarrow & X'' \\ \downarrow & & \downarrow & & \downarrow \\ Y' & \longrightarrow & Y & \longrightarrow & Y'' \\ \downarrow & & \downarrow & & \downarrow \\ Z' & \longrightarrow & Z & \longrightarrow & Z'' \end{array}$$

such that $X' \rightarrow Y' \rightarrow Z'$ is a cofiber sequence.

Proof. (1) and (2) are from the universal property of cofibers. Besides, (1) and (2) give the existence of functors $\Delta^2 \times \Delta^2 \rightarrow \mathcal{C}$ such that all rows are cofiber sequences. The last column is cofiber sequence is because $\text{Ho}(\mathcal{C})$ is a triangulated category whose distinguished triangles are cofiber sequences [Lur17, Theorem 1.1.2.14]. $\square$

Lemma 1.6. [AGH19, Lemma 2.21] Given a functor $\Delta^1 \times \Delta^1 \rightarrow \mathcal{C}$ represented by the diagram

$$\begin{array}{ccc} M & \xrightarrow{f} & N \\ \downarrow & & \downarrow \\ P & \xrightarrow{g} & Q \end{array},$$

the induced map $\text{cofib}(f) \rightarrow \text{cofib}(g)$ is an equivalence if and only if this square is a pushout square.

Lemma 1.7. [Lur09, Lemma 4.4.2.1] Let $\mathcal{C}$ be an ∞ -category and suppose we are given a functor $\Delta^2 \times \Delta^1 \rightarrow \mathcal{C}$ which we will depict as a diagram

$$\begin{array}{ccccc} X & \longrightarrow & Y & \longrightarrow & Z \\ \downarrow & & \downarrow & & \downarrow \\ X' & \longrightarrow & Y' & \longrightarrow & Z' \end{array}$$

Suppose that the left square is a pushout in $\mathcal{C}$. Then the right square is a pushout if and only if the outer square is a pushout.

Definition 1.8. An *exact* functor between stable ∞ -categories is a functor preserving zero objects and fiber sequences.

Proposition 1.9. [Lur17, Proposition 1.1.4.1]

A functor between stable ∞ -categories is exact if and only if it preserves finite limits and colimits.

The collection $\text{Cat}_\infty^{\text{ex}} \subseteq \text{Cat}_\infty$ of stable ∞ -categories and exact functors is a subcategory.

Theorem 1.10. [Lur17, Theorem 1.1.4.4, Proposition 1.1.4.6] The ∞ -category $\text{Cat}_\infty^{\text{ex}}$ admits all finite limits and filtered colimits.

Definition 1.11. An ∞ -functor between ∞ -categories is called *left exact* if it preserves all finite limits.

Lemma 1.12. [Lur09, Corollary] An ∞ -functor $F : \mathcal{C} \rightarrow \mathcal{D}$ out of an ∞ -category $\mathcal{C}$ that has all finite colimits is left exact if and only if it preserves pullbacks and the terminal object.

An ∞ -category is *idempotent-complete* if the image of this category under the Yoneda embedding is closed with respect to retracts. We denote the category of small idempotent-complete stable ∞ -categories as $\text{Cat}_\infty^{\text{perf}}$. The inclusion $\text{Cat}_\infty^{\text{perf}} \rightarrow \text{Cat}_\infty^{\text{ex}}$ has a left adjoint, referred to as $\text{Idem} : \text{Cat}_\infty^{\text{ex}} \rightarrow \text{Cat}_\infty^{\text{perf}}$, which is known as *idempotent completion* [Lur09]. Hence, the inclusion $\text{Cat}_\infty^{\text{perf}} \rightarrow \text{Cat}_\infty^{\text{ex}}$ is left exact.

1.2 Localizing invariants

Definition 1.13. [BGT13, Definition 5.12, Proposition 5.15]

An *exact sequence* of small stable idempotent-complete ∞ -categories $\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{C}$ is a sequence where the composite morphism is zero, the functor $\mathcal{A} \rightarrow \mathcal{B}$ is fully faithful, and the morphism from $\mathcal{B}/\mathcal{A}$ to $\mathcal{C}$ constitutes an equivalence after idempotent-completion.

Definition 1.14. [BGT13, Definition 5.18]

A *split-exact* sequence of small stable ∞ -categories is an exact sequence $\mathcal{A} \xrightarrow{f} \mathcal{B} \xrightarrow{g} \mathcal{C}$ with exact functors $i : \mathcal{B} \rightarrow \mathcal{A}$ and $j : \mathcal{C} \rightarrow \mathcal{B}$, which serve as right adjoints to f and g , respectively, such that $i \circ f \simeq \text{Id}$ and $g \circ j \simeq \text{Id}$.

Definition 1.15. [BGT13, Definition 2.14]

Two small stable ∞ -categories are called *Morita equivalent* if their idempotent completions are equivalent.

Definition 1.16. A functor $F : \text{Cat}_\infty^{\text{ex}} \rightarrow \mathcal{D}$ into a stable presentable ∞ -category $\mathcal{D}$ is a *Morita invariant* if it sends Morita equivalences to homotopy equivalences.

Definition 1.17. [BGT13, Definition 6.1]

A functor $F : \text{Cat}_\infty^{\text{perf}} \rightarrow \mathcal{D}$ into a stable presentable ∞ -category $\mathcal{D}$ is an *additive invariant* if

1. It preserves zero objects.
2. It commutes with filtered colimits.
3. it sends split-exact sequence

$$\mathcal{A} \begin{array}{c} \xrightarrow{f} \\ \xleftarrow{i} \end{array} \mathcal{B} \begin{array}{c} \xrightarrow{g} \\ \xleftarrow{j} \end{array} \mathcal{C}$$

of small stable ∞ -categories to an equivalence $F(\mathcal{A}) \vee F(\mathcal{B}) \xrightarrow{\sim} F(\mathcal{C})$ in $\mathcal{D}$.

Definition 1.18. [BGT13, Definition 8.1]

A functor $F : \text{Cat}_\infty^{\text{perf}} \rightarrow \mathcal{D}$ into a stable presentable ∞ -category $\mathcal{D}$ is a *localizing invariant* if

1. It preserves zero objects.
2. It commutes with filtered colimits.
3. it sends exact sequence

$$\mathcal{A} \rightarrow \mathcal{B} \rightarrow \mathcal{C}$$

of small stable idempotent-complete ∞ -categories to fiber sequences

$$F(\mathcal{A}) \rightarrow F(\mathcal{B}) \rightarrow F(\mathcal{C})$$

in $\mathcal{D}$.

Remark 1.19. A localizing invariant is always an additive invariant.

Example 1.20. [BGT13] constructed the universal additive invariant and localizing invariant:

$$\mathcal{U}_{\text{add}} : \text{Cat}_{\infty}^{\text{perf}} \rightarrow \mathcal{M}_{\text{add}}, \mathcal{U}_{\text{loc}} : \text{Cat}_{\infty}^{\text{perf}} \rightarrow \mathcal{M}_{\text{loc}}.$$

[BGT13, Section 5] construct the connective K-theory $K^{\text{cn}} : \text{Cat}_{\infty}^{\text{ex}} \rightarrow \text{Sp}$ of small stable ∞ -categories. Its restriction on $\text{Cat}_{\infty}^{\text{perf}}$ is an additive invariant. It is not a Morita invariant. The non-connective K-theory $K : \text{Cat}_{\infty}^{\text{ex}} \rightarrow \text{Sp}$ of small stable ∞ -categories, which is constructed by [BGT13, Section 7], is a Morita invariant. Besides, its restriction on $\text{Cat}_{\infty}^{\text{perf}}$ is a localizing invariant.

There are many other differential constructions of higher algebraic K-theory. Waldhausen[Wal85] constructed algebraic K-theory for Waldhausen categories. Also, algebraic K-theory are defined for model categories of simplicial categories and dg categories. However, they are very difficult to compute. One of the reason we choose the algebraic K-theory of ∞ -categories is easy. Except for [BGT13], the K-theory of ∞ -categories is also studied by [Bar16]. Other reason is because it very extensive. [RSW24] shows every spectrum is the k-theory of a stable ∞ -category. So we except that it covers any other constructions of algebraic K-theory.

2 DÉVISSAGE CONDITION FOR EXACT FUNCTORS

2.1 Category of sequences

Definition 2.1. [BGT13, Definition 7.1] Let $\text{Gap}([n], \mathcal{C})$ be the full subcategory of $\text{Fun}(\text{N}(\text{Ar}[n]), \mathcal{C})$ spanned by the functors $X_{\bullet, \bullet} : \text{N}(\text{Ar}[n]) \rightarrow \mathcal{C}$ such that, for each $i \in I$, $X_{i,i}$ is a zero object of $\mathcal{C}$, and for each $i < j < k$, the square

$$\begin{array}{ccc} X_{i,j} & \longrightarrow & X_{i,k} \\ \downarrow & & \downarrow \\ X_{j,j} & \longrightarrow & X_{j,k} \end{array}$$

is pushout.

Lemma 2.2. [BGT13, Lemma 7.3] Let $\mathcal{C}$ be a stable ∞ -category. The forgetful functor

$$\text{Gap}([n], \mathcal{C}) \rightarrow \text{Fun}(\Delta^{n-1}, \mathcal{C}), X_{\bullet, \bullet} \mapsto X_{0, \bullet}$$

is an equivalence of ∞ -category.

Let $F_{\bullet, \bullet} : X_{\bullet, \bullet} \rightarrow Y_{\bullet, \bullet}$ be a morphism in $\text{Gap}([n], \mathcal{C})$, then $\text{fib}(F_{\bullet, \bullet})_{i,j} \simeq \text{fib}(F_{i,j})$ and $\text{cofib}(F_{\bullet, \bullet})_{i,j} \simeq \text{cofib}(F_{i,j})$. Consider the functor $[n+1] \rightarrow [n]$ which sends $i \mapsto i$ (if $i \leq n$) and $n+1 \mapsto n$. This defines a functor:

$$\text{Ar}[n+1] \rightarrow \text{Ar}[n]$$

which induces an exact inclusion functor of stable ∞ -categories

$$i_n : \text{Gap}([n], \mathcal{C}) \hookrightarrow \text{Gap}([n+1], \mathcal{C}).$$

Define

$$\text{Gap}(\mathcal{C}) := \text{colim} \left\{ \mathcal{C} \xrightarrow{i_1} \text{Gap}([2], \mathcal{C}) \xrightarrow{i_2} \cdots \xrightarrow{i_n} \text{Gap}([n+1], \mathcal{C}) \xrightarrow{i_{n+1}} \cdots \right\}.$$

Define by $\text{Gap}^w([n], \mathcal{C})$ the full subcategory of $\text{Gap}([n], \mathcal{C})$ whose objects are $X_{\bullet, \bullet}$ with $X_{0,n} = 0$ an zero object in $\mathcal{C}$. Thus, the exact inclusion i_n restricts to an exact inclusion

$$i_n^w : \text{Gap}^w([n], \mathcal{C}) \hookrightarrow \text{Gap}^w([n+1], \mathcal{C}).$$

Define

$$\text{Gap}^w(\mathcal{C}) := \text{colim} \left\{ 0 \xrightarrow{i_1^w} \text{Gap}^w([2], \mathcal{C}) \xrightarrow{i_2^w} \cdots \xrightarrow{i_n^w} \text{Gap}^w([n+1], \mathcal{C}) \xrightarrow{i_{n+1}^w} \cdots \right\}.$$

Definition 2.3. Assume $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C}$ is an exact functor of small stable ∞ -categories. Define $\text{Gap}([n], \mathcal{F})$ and $\text{Gap}^w([n], \mathcal{F})$ by the pullbacks

$$\begin{array}{ccc} \text{Gap}([n], \mathcal{F}) & \longrightarrow & \text{Gap}([n], \mathcal{C}) \\ \downarrow & & \downarrow \\ \mathcal{A}^n & \xrightarrow{\mathcal{F}^n} & \mathcal{C}^n \end{array}, \quad \begin{array}{ccc} \text{Gap}^w([n], \mathcal{F}) & \longrightarrow & \text{Gap}^w([n], \mathcal{C}) \\ \downarrow & & \downarrow \\ \mathcal{A}^n & \xrightarrow{F^n} & \mathcal{C}^n \end{array}$$

where $\text{Gap}([n], \mathcal{C}) \rightarrow \mathcal{C}^n$ is defined by $X_{\bullet, \bullet} \mapsto (X_{0,1}, \dots, X_{n-1,n})$ and $\text{Gap}^w([n], \mathcal{C}) \rightarrow \mathcal{C}^n$ is its restriction.

$\text{Gap}([n], \mathcal{F})$ consists of

- Objects are pair $(A_{\bullet}, X_{\bullet, \bullet})$, where $A_{\bullet} \in \mathcal{A}^n, X_{\bullet, \bullet} \in \text{Gap}([n], \mathcal{C})$ are objects such that $\mathcal{F}(A_i) = X_{i-1,i}$.
- Morphisms are pair $(f_{\bullet}, F_{\bullet, \bullet})$, where $f_{\bullet} \in \mathcal{A}^n, F_{\bullet, \bullet} \in \text{Gap}([n], \mathcal{C})$ are morphisms such that $\mathcal{F}(f_i) = F_{i-1,i}$.

Thus, the exact inclusion i_n and $\mathcal{A}^n \rightarrow \mathcal{A}^{n+1}, (A_1, \dots, A_n) \mapsto (A_1, \dots, A_n, 0)$ induces an exact inclusion

$$i_{n, \mathcal{F}} : \text{Gap}([n], \mathcal{F}) \hookrightarrow \text{Gap}([n+1], \mathcal{F})$$

Define

$$\text{Gap}(\mathcal{F}) := \text{colim} \left\{ \mathcal{A} \xrightarrow{i_{1, \mathcal{F}}} \text{Gap}([2], \mathcal{F}) \xrightarrow{i_{2, \mathcal{F}}} \dots \xrightarrow{i_{n, \mathcal{A}}^w} \text{Gap}^w([n+1], \mathcal{F}) \xrightarrow{i_{n+1, \mathcal{F}}} \dots \right\}.$$

$\text{Gap}^w([n], \mathcal{F})$ is naturally a full subcategory of $\text{Gap}([n], \mathcal{F})$. Then, the exact inclusion $i_{n, \mathcal{A}}$ restricts to an exact inclusion

$$i_{n, \mathcal{F}}^w : \text{Gap}^w([n], \mathcal{F}) \hookrightarrow \text{Gap}^w([n+1], \mathcal{F}).$$

Define

$$\text{Gap}^w(\mathcal{F}) := \text{colim} \left\{ 0 \xrightarrow{i_{1, \mathcal{F}}^w} \text{Gap}^w([2], \mathcal{F}) \xrightarrow{i_{2, \mathcal{F}}^w} \dots \xrightarrow{i_{n, \mathcal{F}}^w} \text{Gap}^w([n+1], \mathcal{F}) \xrightarrow{i_{n+1, \mathcal{F}}^w} \dots \right\}.$$

$\text{Gap}([n], \mathcal{F}), \text{Gap}^w([n], \mathcal{F})$ and $\text{Gap}(\mathcal{F}), \text{Gap}^w(\mathcal{F})$ are stable. Furthermore, if $\mathcal{C}$ and $\mathcal{A}$ are idempotent-complete, so do them. $\text{Gap}, \text{Gap}^w : \mathcal{F} \mapsto \text{Gap}(\mathcal{F}), \text{Gap}^w(\mathcal{F})$ can be seen as functors $\text{Fun}(\Delta^1, \text{Cat}_{\infty}^{\text{ex}}) \rightarrow \text{Cat}_{\infty}^{\text{ex}}$.

2.2 Definition of dévissage condition

Proposition 2.4. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{ex}}$ be an exact functor between small stable ∞ -categories. Then the following are equivalent:

- (1) For any morphism $f : X \rightarrow Y$ in $\mathcal{C}$, there is a factorization

$$f : X = X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_n = Y$$

such that $\text{cofib}(X_i \rightarrow X_{i+1}) \in \mathcal{F}(\mathcal{A})$ up to equivalences.

- (2) for any morphism $f : 0 \rightarrow Y$ in $\mathcal{C}$, there is a factorization

$$f : 0 = X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_n = Y$$

such that $\text{cofib}(X_i \rightarrow X_{i+1}) \in \mathcal{F}(\mathcal{A})$ up to equivalences.

- (3) For any morphism $f : X \rightarrow 0$ in $\mathcal{C}$, there is a factorization

$$f : X = X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_n = 0$$

such that $\text{cofib}(X_i \rightarrow X_{i+1}) \in \mathcal{F}(\mathcal{A})$ up to equivalences.

Proof. (1) $\Rightarrow$ (2), (3) is obvious. Now we prove (2) $\Rightarrow$ (1) and (3) $\Rightarrow$ (1).

(2) $\Rightarrow$ (1). For a morphism $X \rightarrow Y$, assume Z is its cofiber. From (2), there is a sequence

$$0 = Z_0 \rightarrow \cdots \rightarrow Z_n = Z$$

such that $\text{cofib}(Z_i \rightarrow Z_{i+1}) \in \mathcal{F}(\mathcal{A})$ up to equivalences. Define X_i by pullback

$$\begin{array}{ccc} X_i & \longrightarrow & Y \\ \downarrow & & \downarrow \\ Z_i & \longrightarrow & Z \end{array}$$

Then $X_n = Y$ and $X_0 = \text{fib}(Y \rightarrow Z) = X$. There are also pushout squares in a stable ∞ -category. Hence, we get pushout squares

$$\begin{array}{ccc} X_i & \longrightarrow & X_{i+1} \\ \downarrow & & \downarrow \\ Z_i & \longrightarrow & Z_{i+1} \end{array},$$

by Lemma 1.7. By Lemma 1.6, we get $\text{cofib}(X_i \rightarrow X_{i+1}) \simeq \text{cofib}(Z_i \rightarrow Z_{i+1}) \in \mathcal{F}(\mathcal{A})$ up to equivalences. Then the sequence

$$X = X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_n = Y$$

is what we desired.

(3) $\Rightarrow$ (1). For a morphism $X \rightarrow Y$, assume W is its fiber. From (3), there is a sequence

$$W = W_0 \rightarrow \cdots \rightarrow W_n = W$$

such that $\text{cofib}(W_i \rightarrow W_{i+1}) \in \mathcal{F}(\mathcal{A})$. Define X_i by pushout

$$\begin{array}{ccc} W & \longrightarrow & X \\ \downarrow & & \downarrow \\ W_i & \longrightarrow & X_i \end{array}$$

Then $X_0 = X$ and $X_n = \text{cofib}(W \rightarrow X) = Y$. Then we get pushout squares

$$\begin{array}{ccc} W_i & \longrightarrow & X_i \\ \downarrow & & \downarrow \\ W_{i+1} & \longrightarrow & X_{i+1} \end{array}.$$

by Lemma 1.7. By Lemma 1.6, we get $\text{cofib}(X_i \rightarrow X_{i+1}) \simeq \text{cofib}(W_i \rightarrow W_{i+1}) \in \mathcal{F}(\mathcal{A})$ up to equivalences. Then the sequence

$$X = X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_n = Y$$

is what we desired. □

Remark 2.5. We will omit "up to equivalence" in the following.

Definition 2.6. We say the exact functor $\mathcal{F}$ satisfies the *dévissage condition* if the equivalent descriptions in the Proposition 2.4 hold true.

Example 2.7. Let $\mathcal{C}$ be a stable ∞ -category. Then $\text{id}_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}$ and $0_{\mathcal{C}} : \mathcal{C} \rightarrow *$ satisfy the dévissage condition.

Example 2.8. A fully faithful functor satisfies the dévissage condition if and only if it is an equivalence.

Denote by $E(\mathcal{F})$ the full subcategory of $\mathcal{C}$ consisting of objects X such that $0 \rightarrow X$ admits a factorization as above, i.e. the extension closure of objects in $\mathcal{F}(\mathcal{A})$. By [Lur17, Lemma 1.1.3.3], $E(\mathcal{F})$ is stable full subcategory of $\mathcal{C}$. We can naturally get the following results.

Proposition 2.9. $\mathcal{F}$ satisfies the dévissage condition if and only if $E(\mathcal{F}) = \mathcal{C}$.

Remark 2.10. Ofcourse, $\mathcal{A} \rightarrow E(\mathcal{F})$ satisfies the dévissage condition.

When we assume $\mathcal{C}$ and $\mathcal{A}$ are idempotent-complete, we only need the weaker condition in some cases.

Definition 2.11. Assume $\mathcal{C}$ and $\mathcal{A}$ are idempotent-complete. We say the $\mathcal{F}$ satisfies the *weak dévissage condition* if $\text{Idem}(E(\mathcal{F})) = \mathcal{C}$.

Remark 2.12. $\text{Idem}(E(\mathcal{F}))$ is just the thick closure of $\text{Ob}(\mathcal{F}(\mathcal{A}))$ in $\mathcal{C}$, i.e. the smallest stable full subcategory of $\mathcal{C}$ containing $\mathcal{F}(\mathcal{A})$ which is closed under extensions and direct factors.

There are exact “evaluation” functors:

$$\text{ev}_{n,\mathcal{A}} : \text{Gap}([n], \mathcal{F}) \rightarrow \mathcal{C}, (A_\bullet, X_{\bullet,\bullet}) \mapsto X_{0,n}.$$

These are compatible with stabilization along the inclusion functors $i_{n,\mathcal{F}}$, so there is also an induced exact functor:

$$\text{ev}_{\mathcal{A}} : \text{Gap}(\mathcal{F}) \rightarrow \mathcal{C}.$$

Proposition 2.13. The functor $\text{ev}_{\mathcal{F}} : \text{Gap}(\mathcal{F}) \rightarrow \mathcal{C}$ induces a well-defined functor

$$\widehat{\text{ev}}_{\mathcal{F}} : \text{Gap}(\mathcal{F}) / \text{Gap}^w(\mathcal{F}) \rightarrow E(\mathcal{F}),$$

which is an equivalence.

Proof. By [BGT13, Corollary 5.11, Proposition 5.15], we only need to prove

$$\text{Ho}(\text{Gap}(\mathcal{F})) / \text{Ho}(\text{Gap}^w(\mathcal{F})) \rightarrow \text{Ho}(E(\mathcal{F}))$$

is an equivalence of homotopy categories. We prove it by steps.

Step 1: For every morphism $\text{ev}_{\mathcal{F}}(A_\bullet, X_{\bullet,\bullet}) \rightarrow Y$ in $E(\mathcal{F})$, there is a morphism $(A_\bullet, X_{\bullet,\bullet}) \rightarrow (A'_\bullet, X'_{\bullet,\bullet})$ such that $\text{ev}_{\mathcal{F}}((A_\bullet, X_{\bullet,\bullet}) \rightarrow (A'_\bullet, X'_{\bullet,\bullet})) \simeq \text{ev}_{\mathcal{F}}(A'_\bullet, X'_{\bullet,\bullet}) \rightarrow Y$.

Assume $(A_\bullet, X_{\bullet,\bullet}) \in \text{Gap}([n], \mathcal{F})$. Since $\mathcal{A} \rightarrow E(\mathcal{F})$ satisfies the dévissage condition, there is a factorization

$$\text{ev}_{\mathcal{F}}(A_\bullet, X_{\bullet,\bullet}) = Y_0 \rightarrow Y_1 \rightarrow \cdots \rightarrow Y_m = Y$$

such that $\text{cofib}(Y_i \rightarrow Y_{i+1}) \in \mathcal{F}(\mathcal{A})$. Assume $B_1, \dots, B_m \in \mathcal{A}$ such that $\mathcal{F}(B_i) = \text{cofib}(Y_{i-1} \rightarrow Y_i)$. Then $(A_\bullet, X_{\bullet,\bullet}) \rightarrow (A'_\bullet, X'_{\bullet,\bullet})$ is defined as follows

$$\begin{array}{c} X_{\bullet,\bullet} : \quad 0 \longrightarrow X_{0,1} \longrightarrow \cdots \longrightarrow X_{0,n} \longrightarrow X_{0,n} \longrightarrow \cdots \longrightarrow X_{0,n} \\ \quad \quad \downarrow \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \downarrow \quad \quad \quad \downarrow \\ X'_{\bullet,\bullet} : \quad 0 \longrightarrow X_{0,1} \longrightarrow \cdots \longrightarrow X_{0,n} \longrightarrow Y_1 \longrightarrow \cdots \longrightarrow Y_n \\ \\ A_\bullet : \quad A_1 \quad \quad \cdots \quad \quad A_n \quad \quad 0 \quad \quad \cdots \quad \quad 0 \\ \quad \quad \downarrow \text{id} \quad \quad \quad \downarrow \text{id} \quad \quad \downarrow 0 \quad \quad \quad \downarrow 0 \\ A'_\bullet : \quad A_1 \quad \quad \cdots \quad \quad A_n \quad \quad B_1 \quad \quad \cdots \quad \quad B_m \end{array}$$

It is well-defined since $X_{0,n} = Y_0$.

Step 2: If $\text{ev}_{\mathcal{F}}(A_{\bullet}, X_{\bullet, \bullet}) \simeq \text{ev}_{\mathcal{F}}(B_{\bullet}, Y_{\bullet, \bullet})$, $(A_{\bullet}, X_{\bullet, \bullet}) \simeq (B_{\bullet}, Y_{\bullet, \bullet})$ in $\text{Gap}(\mathcal{F})/\text{Gap}^w(\mathcal{F})$.

Assume $(A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet}) \in \text{Gap}([n], \mathcal{F})$. Then define an object $(C_{\bullet}, Z_{\bullet, \bullet})$ and two morphisms $(C_{\bullet}, Z_{\bullet, \bullet}) \rightarrow (A_{\bullet}, X_{\bullet, \bullet})$ and $(C_{\bullet}, Z_{\bullet, \bullet}) \rightarrow (B_{\bullet}, Y_{\bullet, \bullet})$ as follows.

$$\begin{array}{c}
 X_{\bullet, \bullet} : \quad 0 \longrightarrow \cdots \longrightarrow X_{0,n} \longrightarrow X_{0,n} \longrightarrow \cdots \longrightarrow X_{0,n} \\
 \quad \quad \quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow \\
 Z_{\bullet, \bullet} : \quad 0 \longrightarrow \cdots \longrightarrow 0 \longrightarrow X_{0,1} \longrightarrow \cdots \longrightarrow X_{0,n} \\
 \quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \\
 Y_{\bullet, \bullet} : \quad 0 \longrightarrow \cdots \longrightarrow Y_{0,n} \longrightarrow Y_{0,n} \longrightarrow \cdots \longrightarrow Y_{0,n}
 \end{array}$$

$$\begin{array}{c}
 A_{\bullet} : \quad A_1 \quad \quad \cdots \quad \quad A_n \quad \quad 0 \quad \quad \cdots \quad \quad 0 \\
 \quad \quad \uparrow \quad \quad \quad \uparrow \quad \quad \uparrow \quad \quad \quad \uparrow \\
 C_{\bullet} : \quad 0 \quad \quad \cdots \quad \quad 0 \quad \quad A_1 \quad \quad \cdots \quad \quad A_n \\
 \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \downarrow \quad \quad \quad \downarrow \\
 B_{\bullet} : \quad B_1 \quad \quad \cdots \quad \quad B_n \quad \quad 0 \quad \quad \cdots \quad \quad 0
 \end{array}$$

There are well-defined since $X_{0,n} \simeq Y_{0,n}$. One can check that there are equivalences in $\text{Gap}(\mathcal{F})/\text{Gap}^w(\mathcal{F})$.

Step 3: For two morphisms $(f_{\bullet}, F_{\bullet, \bullet}), (g_{\bullet}, G_{\bullet, \bullet}) : (A_{\bullet}, X_{\bullet, \bullet}) \rightarrow (B_{\bullet}, Y_{\bullet, \bullet})$, if $\text{ev}_{\mathcal{F}}(f_{\bullet}, F_{\bullet, \bullet}) \simeq \text{ev}_{\mathcal{F}}(g_{\bullet}, G_{\bullet, \bullet})$, then $(f_{\bullet}, F_{\bullet, \bullet}) \simeq (g_{\bullet}, G_{\bullet, \bullet})$ in $\text{Gap}(\mathcal{F})/\text{Gap}^w(\mathcal{F})$.

Assume $(f_{\bullet}, F_{\bullet, \bullet}), (g_{\bullet}, G_{\bullet, \bullet}) \in \text{Gap}([n], \mathcal{F})$. Consider the diagram

$$\begin{array}{c}
 X'_{\bullet, \bullet} : \quad 0 \longrightarrow \cdots \longrightarrow 0 \xrightarrow{F_{0,1}} X_{0,1} \longrightarrow \cdots \xrightarrow{F_{0,n}} X_{0,n} \\
 \quad \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \\
 Y'_{\bullet, \bullet} : \quad 0 \xrightarrow{\quad} \cdots \longrightarrow 0 \xrightarrow{\quad} Y_{0,1} \longrightarrow \cdots \longrightarrow Y_{0,n} \\
 \quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \\
 X_{\bullet, \bullet} : \quad 0 \longrightarrow \cdots \longrightarrow X_{0,n} \longrightarrow X_{0,n} \longrightarrow \cdots \longrightarrow X_{0,n} \\
 \quad \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \\
 Y_{\bullet, \bullet} : \quad 0 \longrightarrow \cdots \longrightarrow Y_{0,n} \longrightarrow Y_{0,n} \longrightarrow \cdots \longrightarrow Y_{0,n}
 \end{array}$$

$$\begin{array}{c}
 A'_{\bullet} : \quad 0 \quad \quad \cdots \quad \quad 0 \quad \quad A_1 \quad \quad \cdots \quad \quad A_n \\
 \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \\
 B'_{\bullet} : \quad 0 \quad \quad \cdots \quad \quad 0 \quad \quad B_1 \quad \quad \cdots \quad \quad B_n \\
 \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \downarrow \quad \quad \quad \downarrow \\
 A_{\bullet} : \quad A_1 \quad \quad \cdots \quad \quad A_n \quad \quad 0 \quad \quad \cdots \quad \quad 0 \\
 \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \quad \quad \swarrow \downarrow \quad \quad \quad \swarrow \downarrow \\
 B_{\bullet} : \quad B_1 \quad \quad \cdots \quad \quad B_n \quad \quad 0 \quad \quad \cdots \quad \quad 0
 \end{array}$$

Here $(A'_{\bullet}, X'_{\bullet, \bullet}) \rightarrow (A_{\bullet}, X_{\bullet, \bullet})$ and $(B'_{\bullet}, Y'_{\bullet, \bullet}) \rightarrow (B_{\bullet}, Y_{\bullet, \bullet})$ are just identity in $\text{Gap}(\mathcal{F})/\text{Gap}^w(\mathcal{F})$ and $(A_{\bullet}, X_{\bullet, \bullet}) \rightarrow (B_{\bullet}, Y_{\bullet, \bullet})$ can be either $(f_{\bullet}, F_{\bullet, \bullet})$ or $(g_{\bullet}, G_{\bullet, \bullet})$. Hence, $(f_{\bullet}, F_{\bullet, \bullet}), (g_{\bullet}, G_{\bullet, \bullet})$ are both equivalent to $(A'_{\bullet}, X'_{\bullet, \bullet}) \rightarrow (B'_{\bullet}, Y'_{\bullet, \bullet})$ in $\text{Gap}(\mathcal{F})/\text{Gap}^w(\mathcal{F})$.

Step 4: $\text{Ho}(\widehat{\text{ev}}_{\mathcal{F}}) : \text{Ho}(\text{Gap}(\mathcal{F}))/\text{Ho}(\text{Gap}^w(\mathcal{F})) \rightarrow \text{Ho}(\text{E}(\mathcal{F}))$ is an equivalence.

Step 1 tells us it is essentially surjective. Step 1 and Step 2 tells us it is surjective on morphisms. Step 3 tells us it is injective on morphisms. Hence, it is both essentially surjective and fully faithful.

□

Corollary 2.14. $\mathcal{F}$ satisfies the dévissage condition if and only if $\text{ev}_{\mathcal{F}}$ is essentially surjective.

3 THEOREMS FOR INVARIANTS

3.1 Additivity theorem

Assume $\mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ is a small idempotent-complete stable ∞ -category and $\mathcal{F}_{\mathcal{A}} : \mathcal{A} \rightarrow \mathcal{C}, \mathcal{F}_{\mathcal{B}} : \mathcal{B} \rightarrow \mathcal{C}$ are exact functors between idempotent-complete stable ∞ -categories. Define a category $\mathcal{S}(\mathcal{F}_{\mathcal{A}}, \mathcal{F}_{\mathcal{B}})$ by

pullback

$$\begin{array}{ccc} \mathcal{S}(\mathcal{F}_{\mathcal{A}}, \mathcal{F}_{\mathcal{B}}) & \longrightarrow & \text{Gap}([2], \mathcal{C}) \\ \downarrow & & \downarrow \\ \mathcal{A} \times \mathcal{B} & \xrightarrow{(\mathcal{F}_{\mathcal{A}}, \mathcal{F}_{\mathcal{B}})} & \mathcal{C}^2 \end{array}$$

Where $\text{Gap}([2], \mathcal{C}) \rightarrow \mathcal{C}$ is defined by $X_{\bullet, \bullet} \mapsto (X_{0,1}, X_{1,2})$. Then $\mathcal{S}(\mathcal{F}_{\mathcal{A}}, \mathcal{F}_{\mathcal{B}})$ is also stable and idempotent-complete.

Theorem 3.1 (∞ -categorical Additivity Theorem).

Assume $\mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ is a small idempotent-complete stable ∞ -category and $\mathcal{F}_{\mathcal{A}} : \mathcal{A} \rightarrow \mathcal{C}, \mathcal{F}_{\mathcal{B}} : \mathcal{B} \rightarrow \mathcal{C}$ are exact functors between idempotent-complete stable ∞ -categories. The natural projection $\mathcal{S}(\mathcal{F}_{\mathcal{A}}, \mathcal{F}_{\mathcal{B}}) \rightarrow \mathcal{A} \times \mathcal{B}$ induces a homotopy equivalence $\mathcal{U}_{\text{add}}(\mathcal{S}(\mathcal{F}_{\mathcal{A}}, \mathcal{F}_{\mathcal{B}})) \xrightarrow{\cong} \mathcal{U}_{\text{add}}(\mathcal{A}) \times \mathcal{U}_{\text{add}}(\mathcal{B})$.

Proof. There is a split exact sequence

$$\mathcal{A} \xrightleftharpoons[i]{f} \mathcal{S}(\mathcal{A}, \mathcal{C}, \mathcal{B}) \xrightleftharpoons[j]{g} \mathcal{B},$$

where i, g are projections, f, j are defined by

$$\begin{aligned} f : \mathcal{A} &\rightarrow \mathcal{A} \times \mathcal{B}, X \mapsto (X, 0) \\ \mathcal{A} &\rightarrow \text{Gap}([2], \mathcal{C}), X \mapsto (0 \rightarrow \mathcal{F}_{\mathcal{A}}(X) \rightarrow \mathcal{F}_{\mathcal{A}}(X)) \\ g : \mathcal{B} &\rightarrow \mathcal{A} \times \mathcal{B}, Y \mapsto (0, Y) \\ \mathcal{B} &\rightarrow \text{Gap}([2], \mathcal{C}), Y \mapsto (0 \rightarrow 0 \rightarrow \mathcal{F}_{\mathcal{B}}(Y)). \end{aligned}$$

Thus, it induces a homotopy equivalence

$$\mathcal{U}_{\text{add}}(\mathcal{S}(\mathcal{F}_{\mathcal{A}}, \mathcal{F}_{\mathcal{B}})) \xrightarrow{\cong} \mathcal{U}_{\text{add}}(\mathcal{A}) \times \mathcal{U}_{\text{add}}(\mathcal{B}).$$

□

Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C}$ be an exact functor between small idempotent-complete stable ∞ -categories. Denote by $q_{n, \mathcal{F}} : \text{Gap}([n], \mathcal{F}) \rightarrow \mathcal{A}^n$ the natural projections.

Proposition 3.2. For each $n \geq 1$, the exact functor $q_{n, \mathcal{F}}$ induces a homotopy equivalence:

$$\mathcal{U}_{\text{add}}(\text{Gap}([n], \mathcal{F})) \xrightarrow{\cong} \mathcal{U}_{\text{add}}(\mathcal{A})^n.$$

Furthermore, this also induces a homotopy equivalence $\mathcal{U}_{\text{add}}(\text{Gap}(\mathcal{F})) \simeq \text{colim}_n \mathcal{U}_{\text{add}}(\mathcal{A}^n)$.

Proof. Define $\mathcal{F}_1 : \mathcal{A} \rightarrow \text{Gap}([n], \mathcal{F})$ by

$$\begin{aligned} \mathcal{A} &\rightarrow \text{Gap}([n], \mathcal{C}), X \mapsto (0 \rightarrow \mathcal{F}(X) \rightarrow \mathcal{F}(X) \rightarrow \cdots \rightarrow \mathcal{F}(X)) \\ \mathcal{A} &\rightarrow \mathcal{A}^n, X \mapsto (X, 0, \dots, 0). \end{aligned}$$

Define $\mathcal{F}_2 : \text{Gap}([n-1], \mathcal{F}) \rightarrow \text{Gap}([n], \mathcal{F})$ by

$$\begin{aligned} \text{Gap}([n-1], \mathcal{C}) &\rightarrow \text{Gap}([n], \mathcal{C}), (0 \rightarrow X_{0,1} \rightarrow \cdots \rightarrow X_{0,n-1}) \mapsto (0 \rightarrow 0 \rightarrow X_{0,1} \rightarrow \cdots \rightarrow X_{0,n-1}) \\ \mathcal{A}^{n-1} &\rightarrow \mathcal{A}^n, (A_1, \dots, A_{n-1}) \mapsto (0, A_1, \dots, A_{n-1}) \\ \mathcal{C}^{n-1} &\rightarrow \mathcal{C}^n, (X_1, \dots, X_{n-1}) \mapsto (0, X_1, \dots, X_{n-1}) \end{aligned}$$

These functors admit projections given by the exact functors

$$\begin{aligned} s_1 : \text{Gap}([n], \mathcal{F}) &\rightarrow \mathcal{A}, \quad (A_{\bullet}, X_{\bullet, \bullet}) \mapsto A_1, \\ s_2 : \text{Gap}([n], \mathcal{F}) &\rightarrow \text{Gap}([n-1], \mathcal{F}), \quad (A_{\bullet}, X_{\bullet, \bullet}) \mapsto (0, A_2, \dots, A_n, 0 \rightarrow 0 \rightarrow X_{1,2} \rightarrow \cdots \rightarrow X_{1,n}), \end{aligned}$$

which assemble to an exact functor $\text{Gap}([n], \mathcal{F}) \rightarrow \mathcal{S}(\mathcal{F}_1, \mathcal{F}_2)$. By the Additivity Theorem 3.1, the pair of exact functors

$$(s_1, s_2) : \text{Gap}([n], \mathcal{F}) \xrightarrow{\sim} \text{Gap}([n-1], \mathcal{F}) \times \mathcal{A} : \mathcal{F}_2 \vee \mathcal{F}_2$$

induces inverse homotopy equivalences for $\mathcal{U}_{\text{add}}$. Inductively, this yields homotopy equivalences

$$\mathcal{U}_{\text{add}}(\text{Gap}([n], \mathcal{F})) \simeq \mathcal{U}_{\text{add}}(\text{Gap}([n-1], \mathcal{F})) \times \mathcal{U}_{\text{add}}(\mathcal{A}) \simeq \cdots \simeq \mathcal{U}_{\text{add}}(\mathcal{A})^n \simeq \mathcal{U}_{\text{add}}(\mathcal{A}^n).$$

Direct inspection shows the composite homotopy equivalence is defined by the exact functor $q_{n, \mathcal{F}}$. The map $\mathcal{U}_{\text{add}}(q_{n, \mathcal{F}})$ admits a homotopy inverse given by the exact functor

$$\mathcal{A}^n \rightarrow \text{Gap}([n], \mathcal{F}), \quad (A_1, \dots, A_n) \mapsto (A_1, \dots, A_n, 0 \rightarrow \mathcal{F}(A_1) \rightarrow \mathcal{F}(A_1) \vee \mathcal{F}(A_2) \rightarrow \cdots \rightarrow \bigvee_{i=1}^n \mathcal{F}(A_i)).$$

Moreover, we have commutative diagrams of exact functors:

$$\begin{array}{ccc} \text{Gap}([n], \mathcal{F}) & \xrightarrow{i_{n, \mathcal{F}}} & \text{Gap}([n+1], \mathcal{F}) \\ \uparrow \downarrow & & \uparrow \downarrow \\ \mathcal{A}^n & \xrightarrow{\quad} & \mathcal{A}^{n+1} \end{array}$$

where the bottom functor is the canonical inclusion, sending $(A_1, \dots, A_{n-1}) \mapsto (A_1, \dots, A_{n-1}, 0)$. The case $n = \infty$ follows immediately, as $\text{colim}_n \mathcal{U}_{\text{add}}(\text{Gap}([n], \mathcal{F})) \xrightarrow{\sim} \mathcal{U}_{\text{add}}(\text{Gap}(\mathcal{F}))$. $\square$

3.2 Semiorthogonal decompositions

One can refer [Huy06] for the definition of *semiorthogonal decompositions* of triangulated categories. And, [DKSS24] explains the *semiorthogonal decompositions* of stable ∞ -categories.

Definition 3.3. Let $\mathcal{C}$ be a stable ∞ -category. A *semiorthogonal decomposition* is a sequence of stable full subcategories $\mathcal{C}_1, \dots, \mathcal{C}_n$ such that

- (1) For $i > j$ and any $X_i \in \mathcal{C}_i, X_j \in \mathcal{C}_j$, $\text{Map}_{\mathcal{C}}(X_i, X_j)$ is contractible.
- (2) For any $X \in \mathcal{C}$, there is a filtration

$$0 = X_n \rightarrow \cdots \rightarrow X_0 = X$$

such that $\text{cofib}(X_i \rightarrow X_{i-1}) \in \mathcal{C}_i$ for any $1 \leq i \leq n$.

We write $\mathcal{C} = \langle \mathcal{C}_1, \dots, \mathcal{C}_n \rangle$.

Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C}$ be an exact functor. Define

$$\mathfrak{p}_{i, \mathcal{C}} : \mathcal{C} \rightarrow \text{Gap}([n], \mathcal{C})$$

by $X \mapsto X_{\bullet, \bullet}$ such that $X_{0, j} = 0$ for $j < i$ and $X_{0, j} \rightarrow X_{0, j+1} = X \xrightarrow{\text{id}} X$ for $j \geq i$.

Lemma 3.4. $\mathfrak{p}_{i, \mathcal{C}}$ is fully faithful and $\mathfrak{p}_{n, \mathcal{C}}(\mathcal{C}), \dots, \mathfrak{p}_{1, \mathcal{C}}(\mathcal{C})$ defines a semiorthogonal decomposition of $\text{Gap}([n], \mathcal{C})$.

Proof. It is easy to see $\mathfrak{p}_{i, \mathcal{C}}$ is fully faithful.

- (1) For $i < j$ and $X, Y \in \mathcal{C}$,

$$\text{Map}_{\text{Gap}([n], \mathcal{C})}(\mathfrak{p}_{i, \mathcal{C}}(X), \mathfrak{p}_{j, \mathcal{C}}(Y)) \simeq \text{Map}_{\text{Fun}(\Delta^1, \mathcal{C})}(X \xrightarrow{\text{id}} X, 0 \rightarrow Y) \simeq 0.$$

(2) For $X_{\bullet,\bullet} \in \text{Gap}([n], \mathbb{C})$, there is a filtration:

$$\begin{array}{ccccccc}
X_{\bullet,\bullet}^0 = 0 : & 0 & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow \cdots \longrightarrow 0 \longrightarrow 0 \\
& \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
X_{\bullet,\bullet}^1 : & 0 & \longrightarrow & X_{0,1} & \longrightarrow & X_{0,1} & \longrightarrow \cdots \longrightarrow X_{0,1} & \longrightarrow & X_{0,1} \\
& \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
& \vdots & & \vdots & & \vdots & & \ddots & & \vdots \\
X_{\bullet,\bullet}^{n-1} : & 0 & \longrightarrow & X_{0,1} & \longrightarrow & X_{0,2} & \longrightarrow \cdots \longrightarrow X_{0,n-1} & \longrightarrow & X_{0,n-1} \\
& \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
X_{\bullet,\bullet}^n = X_{\bullet,\bullet} : & 0 & \longrightarrow & X_{0,1} & \longrightarrow & X_{0,2} & \longrightarrow \cdots \longrightarrow X_{0,n-1} & \longrightarrow & X_{0,n}
\end{array}$$

One can check that $\text{cofib}(X_{\bullet,\bullet}^i \rightarrow X_{\bullet,\bullet}^{i-1}) = \mathfrak{p}_{i,\mathbb{C}}(X_{i-1,i}) \in \mathfrak{p}_{i,\mathbb{C}}(\mathbb{C})$.

□

Define

$$\mathfrak{p}_{i,\mathcal{F}} : \mathcal{A} \rightarrow \text{Gap}([n], \mathcal{F})$$

by $\mathfrak{p}_{i,\mathbb{C}} \circ \mathcal{F} : \mathcal{A} \rightarrow \text{Gap}([n], \mathbb{C})$ and $\mathcal{A} \rightarrow \mathcal{A}^n, A \mapsto A_\bullet$ such that $A_i = A, A_j = 0, \forall j \neq i$.

Proposition 3.5. $\mathfrak{p}_{i,\mathcal{F}}$ is fully faithful and $\mathfrak{p}_{n,\mathcal{F}}(\mathcal{A}), \dots, \mathfrak{p}_{1,\mathcal{F}}(\mathcal{A})$ defines a semiorthogonal decomposition of $\text{Gap}([n], \mathcal{F})$.

Proof. For $X, Y \in \mathcal{A}$,

$$\begin{aligned}
\text{Map}_{\text{Gap}(\mathcal{F})}(\mathfrak{p}_{i,\mathcal{F}}(X), \mathfrak{p}_{i,\mathcal{F}}(Y)) &\simeq \text{Map}_{\text{Gap}(\mathbb{C})}(\mathfrak{p}_{i,\mathbb{C}}(\mathcal{F}(X)), \mathfrak{p}_{i,\mathbb{C}}(\mathcal{F}(Y))) \times_{\text{Map}_{\mathbb{C}}(\mathcal{F}(X), \mathcal{F}(Y))} \text{Map}_{\mathcal{A}}(X, Y) \\
&\simeq \text{Map}_{\mathbb{C}}(\mathcal{F}(X), \mathcal{F}(Y)) \times_{\text{Map}_{\mathbb{C}}(\mathcal{F}(X), \mathcal{F}(Y))} \text{Map}_{\mathcal{A}}(X, Y) \\
&\simeq \text{Map}_{\mathcal{A}}(X, Y).
\end{aligned}$$

Hence, $\mathfrak{p}_{i,\mathcal{F}}$ is fully faithful.

(1) For $i < j$ and $X, Y \in \mathcal{A}$, by Lemma 3.4,

$$\text{Map}_{\text{Gap}(\mathcal{F})}(\mathfrak{p}_{i,\mathcal{F}}(X), \mathfrak{p}_{j,\mathcal{F}}(Y)) = \text{Map}_{\text{Gap}(\mathbb{C})}(\mathfrak{p}_{i,\mathbb{C}}(\mathcal{F}(X)), \mathfrak{p}_{j,\mathbb{C}}(\mathcal{F}(Y))) \times_0 0 = 0.$$

(2) For $(A_\bullet, X_{\bullet,\bullet}) \in \text{Gap}([n], \mathcal{F})$, define $X_{\bullet,\bullet}^i$ from Lemma 3.4 and $A_\bullet^i := (A_1, \dots, A_i, 0, \dots, 0)$. Then we have a sequence

$$(A_\bullet^0, X_{\bullet,\bullet}^0) = 0 \rightarrow \cdots \rightarrow (A_\bullet^n, X_{\bullet,\bullet}^n) = (A_\bullet, X_{\bullet,\bullet})$$

such that $\text{cofib}((A_\bullet^i, X_{\bullet,\bullet}^i) \rightarrow (A_\bullet^{i-1}, X_{\bullet,\bullet}^{i-1})) = \mathfrak{p}_{i,\mathcal{F}}(A_i) \in \mathfrak{p}_{i,\mathcal{F}}(\mathcal{A})$.

□

Remark 3.6. This is a new proof of the equivalence $\mathcal{U}_{\text{add}}(\text{Gap}([n], \mathcal{F})) \simeq \mathcal{U}_{\text{add}}(\mathcal{A})^n$.

3.3 Categorization of fiber with dévissage condition

In this subsection, we still assume $\mathcal{C}$ and $\mathcal{A}$ are idempotent-complete. For each $n \geq 1$, we have an exact functor:

$$q_{n,\mathcal{F}}^w : \text{Gap}^w([n], \mathcal{F}) \rightarrow \mathcal{A}^{n-1}, (A_\bullet, X_\bullet, \bullet) \mapsto (A_1, \dots, A_{n-1}),$$

It has an exact section

$$j_{n,\mathcal{F}} : \mathcal{A}^{n-1} \rightarrow \text{Gap}^w([n], \mathcal{F}),$$

$$(A_1, \dots, A_{n-1}) \mapsto (A_1, \dots, A_{n-1}, \bigvee_{i=1}^{n-1} \Sigma A_i, 0 \rightarrow \mathcal{F}(A_1) \rightarrow \mathcal{F}(A_1) \vee \mathcal{F}(A_2) \rightarrow \dots \rightarrow \bigvee_{i=1}^{n-1} \mathcal{F}(A_i) \rightarrow 0).$$

Define

$$\tau_{n,\mathcal{F}} : \mathcal{A}^{n-1} \rightarrow \text{Gap}^w([n], \mathcal{F}),$$

$$(A_1, \dots, A_{n-1}) \mapsto (A_1, A_2 \vee \Sigma A_1, \dots, A_{n-1} \vee \Sigma A_{n-2}, \Sigma A_{n-1}, 0 \rightarrow \mathcal{F}(A_1) \rightarrow \dots \rightarrow \mathcal{F}(A_{n-1}) \rightarrow 0),$$

where the morphism $\mathcal{F}(A_i) \rightarrow \mathcal{F}(A_{i+1})$ is the zero morphism for each i . The composite functor

$$q_{n,\mathcal{F}}^w \circ \tau_{n,\mathcal{F}} : (A_1, \dots, A_{n-1}) \mapsto (A_1, A_2 \vee \Sigma A_1, \dots, A_{n-1} \vee \Sigma A_{n-2}),$$

induces a homotopy equivalence for any additive invariant.

The diagram

$$\begin{array}{ccc} \text{Gap}^w([n], \mathcal{F}) & \xrightarrow{i_{n,\mathcal{F}}^w} & \text{Gap}^w([n+1], \mathcal{F}) \\ \tau_{n,\mathcal{F}} \uparrow & & \uparrow \tau_{n+1,\mathcal{F}} \\ \mathcal{A}^{n-1} & \xrightarrow{\quad} & \mathcal{A}^n \end{array}$$

commutes, where the bottom functor is the inclusion $(A_1, \dots, A_{n-1}) \mapsto (A_1, \dots, A_{n-1}, 0)$. Thus, we get a functor

$$\tau_{\mathcal{F}} : \text{colim}_n \mathcal{A}^{n-1} \rightarrow \text{Gap}^w(\mathcal{F}).$$

We introduce a result about categorification of cofiber and its proof.

Proposition 3.7. [RSW24, Proposition 3.7] Let $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$ be an exact functor of small stable ∞ -categories. There exists a small stable ∞ -category $\text{Cone}(\mathcal{F})$ with an exact functor $\mathcal{D} \rightarrow \text{Cone}(\mathcal{F})$ that induces an equivalence

$$\text{cofib}(\mathcal{U}_{\text{loc}}(\mathcal{C}) \xrightarrow{\mathcal{U}_{\text{loc}}(\mathcal{F})} \mathcal{U}_{\text{loc}}(\mathcal{D})) \simeq \mathcal{U}_{\text{loc}}(\text{Cone}(\mathcal{F}))$$

Proof. Define the *lex pullback* $\mathcal{D} \times_{\text{Ind}(\mathcal{D})}^{\rightarrow} \text{Ind}(\mathcal{C})$ by the pullback

$$\begin{array}{ccc} \mathcal{D} \times_{\text{Ind}(\mathcal{D})}^{\rightarrow} \text{Ind}(\mathcal{C}) & \longrightarrow & \text{Fun}(\Delta^1, \text{Ind}(\mathcal{D})) \\ \downarrow & & \downarrow \\ \mathcal{D} \times \text{Ind}(\mathcal{C}) & \longrightarrow & \text{Ind}(\mathcal{D}) \times \text{Ind}(\mathcal{D}). \end{array}$$

Then there is a fully faithful functor $\mathcal{C} \rightarrow \mathcal{D} \times_{\text{Ind}(\mathcal{D})}^{\rightarrow} \text{Ind}(\mathcal{C}), X \mapsto (\mathcal{F}(X), X, \mathcal{F}(X) \xrightarrow{\text{id}} \mathcal{F}(X))$. Define $\text{Cone}(\mathcal{F}) := \text{Idem}(\mathcal{D} \times_{\text{Ind}(\mathcal{D})}^{\rightarrow} \text{Ind}(\mathcal{C})/\mathcal{C})$. From [Tam18, Proposition 10], $\mathcal{U}_{\text{loc}}(\mathcal{D} \times_{\text{Ind}(\mathcal{D})}^{\rightarrow} \text{Ind}(\mathcal{C})) \simeq \mathcal{U}_{\text{loc}}(\mathcal{D}) \oplus \mathcal{U}_{\text{loc}}(\text{Ind}(\mathcal{C}))$

$$\text{cofib}(\mathcal{U}_{\text{loc}}(\mathcal{C}) \xrightarrow{\mathcal{U}_{\text{loc}}(\mathcal{F})} \mathcal{U}_{\text{loc}}(\mathcal{D}) \oplus \mathcal{U}_{\text{loc}}(\text{Ind}(\mathcal{C}))) \simeq \mathcal{U}_{\text{loc}}(\text{Cone}(\mathcal{F})).$$

Notice that $\mathcal{U}_{\text{loc}}(\text{Ind}(\mathcal{C})) \simeq 0$ since $\text{Ind}(\mathcal{C})$ admits an Eilenberg swindle. □

Remark 3.8. We replace $\text{Ind}(-)^{\omega_1}$ by $\text{Ind}(-)$. usually, $\text{Ind}(\mathcal{C})$ is not small. To define localizing invariants, we need to use bigger universe. For the details of this technology, readers can refer to [Hen17]. Simply put, we choose two universes $\mathbb{U} \in \mathbb{V}$. Then we use $\text{Ind}^{\mathbb{U}}(\mathcal{C})$, which is $\mathbb{V}$ -small. Hence, we could define localizing invariants of $\text{Ind}^{\mathbb{U}}(\mathcal{C})$. Or, we could use $\text{Ind}(-)^{\kappa}$ for some regular cardinal $\kappa > \omega$ like the construction in the original text [RSW24].

Denote by $Q(\mathcal{F}) := \text{Cone}(\tau_{\mathcal{F}})$. Since $\mathcal{U}_{\text{loc}}(\tau_{\mathcal{F}})$ is the section of $\mathcal{U}_{\text{loc}}(q_{\mathcal{F}}^w)$, the cofiber sequence

$$\mathcal{U}_{\text{loc}}(\text{colim}_n \mathcal{A}^{n-1}) \xrightarrow{\mathcal{U}_{\text{loc}}(\tau_{\mathcal{F}})} \mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{F})) \rightarrow \mathcal{U}_{\text{loc}}(Q(\mathcal{F}))$$

split.

Theorem 3.9. *Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ satisfies the weak dévissage condition, there is a homotopy equivalence*

$$\mathcal{U}_{\text{loc}}(Q(\mathcal{F})) \simeq \text{fib}(\mathcal{U}_{\text{loc}}(\mathcal{F}))$$

Proof. By Proposition 2.13 and the definition of weak dévissage condition, there is a homotopy fiber sequence

$$\mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{F})) \rightarrow \mathcal{U}_{\text{loc}}(\text{Gap}(\mathcal{F})) \rightarrow \mathcal{U}_{\text{loc}}(\mathcal{C}).$$

There is a homotopy fiber sequence

$$\mathcal{U}_{\text{loc}}(\mathcal{A}^{n-1}) \rightarrow \mathcal{U}_{\text{loc}}(\mathcal{A}^n) \rightarrow \mathcal{U}_{\text{loc}}(\mathcal{A})$$

where the left map is induced from $(A_1, \dots, A_{n-1}) \mapsto (A_1, A_2 \vee \Sigma A_1, \dots, A_{n-1} \vee \Sigma A_{n-2}, \Sigma A_{n-1})$ and the right map is induced by $\vee$. It induces a homotopy fiber sequence

$$\mathcal{U}_{\text{loc}}(\text{colim}_n \mathcal{A}^{n-1}) \rightarrow \mathcal{U}_{\text{loc}}(\text{colim}_n \mathcal{A}^n) \rightarrow \mathcal{U}_{\text{loc}}(\mathcal{A})$$

Consider the diagram

$$\begin{array}{ccccc} \mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{F})) & \longrightarrow & \mathcal{U}_{\text{loc}}(\text{Gap}(\mathcal{F})) & \longrightarrow & \mathcal{U}_{\text{loc}}(\mathcal{C}) \\ \mathcal{U}_{\text{loc}}(\tau_{\mathcal{F}}) \uparrow & & \mathcal{U}_{\text{loc}}(\text{colim}_n q_{n,\mathcal{F}}^{-1}) \uparrow & & \mathcal{U}_{\text{loc}}(\mathcal{F}) \uparrow \\ \mathcal{U}_{\text{loc}}(\text{colim}_n \mathcal{A}^{n-1}) & \longrightarrow & \mathcal{U}_{\text{loc}}(\text{colim}_n \mathcal{A}^n) & \longrightarrow & \mathcal{U}_{\text{loc}}(\mathcal{A}) \end{array}$$

Since $\mathcal{U}_{\text{loc}}(\text{colim}_n q_{n,\mathcal{F}}^{-1})$ is a homotopy equivalence by Proposition 3.2, we have a homotopy equivalence by Lemma 1.5

$$\mathcal{U}_{\text{loc}}(Q(\mathcal{F})) \simeq \text{cofib}(\mathcal{U}_{\text{loc}}(\tau_{\mathcal{F}})) \simeq \Omega \text{cofib}(\mathcal{U}_{\text{loc}}(\mathcal{F})).$$

Since $\mathcal{U}_{\text{loc}}$ is into a stable ∞ -category, we get $\mathcal{U}_{\text{loc}}(Q(\mathcal{F})) \simeq \text{fib}(\mathcal{U}_{\text{loc}}(\mathcal{F}))$. □

Remark 3.10. Part of ideas of this Theorem from [Rap18].

Corollary 3.11. *Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. Given a localizing invariant $\mathcal{L}$, if $\mathcal{F}$ satisfies the dévissage condition, $\mathcal{L}(\mathcal{F})$ is a homotopy equivalence if and only if $\mathcal{L}(Q(\mathcal{F}))$ is homotopy trivial.*

Corollary 3.12. *Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. if $\mathcal{F}$ satisfies the dévissage condition and $K_0(\mathcal{F})$ is an isomorphism, $\mathcal{F}$ induces isomorphisms*

$$K_n(\mathcal{F}) : K_n(\mathcal{A}) \xrightarrow{\simeq} K_n(\mathcal{C})$$

for all $n \geq 1$ if and only if $K_n(Q(\mathcal{F})) \simeq 0$ for all $n \geq 0$.

3.4 Categorization of loop

Consider the diagonal map $\delta_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}^2, X \mapsto (X, X)$. It is injective on both objects and morphisms.

Lemma 3.13. $\delta_{\mathcal{C}}$ satisfies the weak dévissage condition.

Proof. For $(X, Y) \in \mathcal{C}^2$, it is a direct factor of

$$(X, Y) \oplus (Y, X) \simeq (X \oplus Y, X \oplus Y) \in \delta_{\mathcal{C}}(\mathcal{C}) \subseteq E(\delta_{\mathcal{C}}).$$

Hence, $(X, Y) \in \text{Idem}(E(\delta_{\mathcal{C}}))$. □

Denote $\theta(\mathcal{C}) = Q(\delta_{\mathcal{C}})$. Then from Theorem 3.9, we get the loop theorem.

Theorem 3.14. Let $\mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be a small stable idempotent-complete ∞ -category. Then we have a homotopy equivalence

$$\mathcal{U}_{\text{loc}}(\theta(\mathcal{C})) \simeq \text{fib}(\mathcal{U}_{\text{loc}}(\delta_{\mathcal{C}})) \simeq \Omega \mathcal{U}_{\text{loc}}(\mathcal{C}).$$

Besides, θ is functorial.

Corollary 3.15. Let $\mathcal{F} : \mathcal{C} \rightarrow \mathcal{D}$ be an exact functor of small stable ∞ -categories. Then there is a homotopy equivalence

$$\mathcal{U}_{\text{loc}}(\text{Cone}(\theta(\mathcal{F}))) \simeq \text{fib}(\mathcal{U}_{\text{loc}}(\mathcal{F})).$$

Let $\mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be a small stable idempotent-complete ∞ -category. Define

$$\text{gr}_n : \text{Gap}([n], \mathcal{C}) \rightarrow \mathcal{C}^n, X_{\bullet, \bullet} \mapsto (X_{0,1}, \dots, X_{n-1,n})$$

and $\text{gr} : \text{Gap}(\mathcal{C}) \rightarrow \text{colim}_n \mathcal{C}^n$ as the colimit $\text{colim}_n \text{gr}_n$. Besides, define some categories of binary sequences by pullbacks

$$\begin{array}{ccc} \text{BGap}([n], \mathcal{C}) & \longrightarrow & \text{Gap}([n], \mathcal{C}) \\ \downarrow & & \downarrow \text{gr}_n \\ \text{Gap}([n], \mathcal{C}) & \xrightarrow{\text{gr}_n} & \mathcal{C}^n \end{array}, \quad \begin{array}{ccc} \text{BGap}^w([n], \mathcal{C}) & \longrightarrow & \text{Gap}^w([n], \mathcal{C}) \\ \downarrow & & \downarrow \text{gr}_n \\ \text{Gap}^w([n], \mathcal{C}) & \xrightarrow{\text{gr}_n} & \mathcal{C}^n \end{array}$$

and

$$\begin{array}{ccc} \text{BGap}(\mathcal{C}) & \longrightarrow & \text{Gap}([n], \mathcal{C}) \\ \downarrow & & \downarrow \text{gr} \\ \text{Gap}(\mathcal{C}) & \xrightarrow{\text{gr}} & \text{colim}_n \mathcal{C}^n \end{array}, \quad \begin{array}{ccc} \text{BGap}^w(\mathcal{C}) & \longrightarrow & \text{Gap}^w(\mathcal{C}) \\ \downarrow & & \downarrow \text{gr} \\ \text{Gap}^w(\mathcal{C}) & \xrightarrow{\text{gr}} & \text{colim}_n \mathcal{C}^n \end{array}.$$

Then the identity define diagonal functors Δ .

Theorem 3.16. [KW19, Theorem 1.1] Let $\mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be a small stable idempotent-complete ∞ -category. There is a homotopy equivalence

$$\text{cofib}(\mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{C}))) \xrightarrow{\mathcal{U}_{\text{loc}}(\Delta)} \mathcal{U}_{\text{loc}}(\text{BGap}^w(\mathcal{C})) \simeq \Omega \mathcal{U}_{\text{loc}}(\mathcal{C}).$$

$\text{BGap}^w(\mathcal{C})$ consisting of

- (1) Objects are 2-tuples $(X_{\bullet, \bullet}, Y_{\bullet, \bullet})$, where $X_{\bullet, \bullet}, Y_{\bullet, \bullet} \in \text{Gap}^w(\mathcal{C})$ and $X_{i-1,i} = Y_{i-1,i}$.
- (2) Morphisms between $(X_{\bullet, \bullet}, Y_{\bullet, \bullet})$ and $(X'_{\bullet, \bullet}, Y'_{\bullet, \bullet})$ are 2-tuples $(f_{\bullet, \bullet} : X_{\bullet, \bullet} \rightarrow X'_{\bullet, \bullet}, g_{\bullet, \bullet} : Y_{\bullet, \bullet} \rightarrow Y'_{\bullet, \bullet})$ of morphisms in $\text{Gap}^w(\mathcal{C})$ such that $g_{i-1,i} \simeq f_{i-1,i}$ for all $i \geq 1$.

And, $\Delta(X_{\bullet, \bullet}) = (X_{\bullet, \bullet}, X_{\bullet, \bullet})$.

Proposition 3.17. Δ satisfies the weak dévissage condition.

Proof. For $(X_{\bullet,\bullet}, Y_{\bullet,\bullet}) \in \mathbf{BGap}^w(\mathcal{C})$, we have

$$(X_{\bullet,\bullet} \oplus Y_{\bullet,\bullet}, Y_{\bullet,\bullet} \oplus X_{\bullet,\bullet}) \simeq (X_{\bullet,\bullet} \oplus Y_{\bullet,\bullet}, X_{\bullet,\bullet} \oplus Y_{\bullet,\bullet}) \in E(\Delta),$$

Then $(X_{\bullet,\bullet}, Y_{\bullet,\bullet})$ is a direct factor of $(X_{\bullet,\bullet} \oplus Y_{\bullet,\bullet}, Y_{\bullet,\bullet} \oplus X_{\bullet,\bullet})$, which means it is in $\text{Idem}(E(\Delta))$. Hence, Δ satisfies the weak dévissage condition. $\square$

Denote $\Theta(\mathcal{C}) := Q(\Delta)$. This define a functor $\Theta : \text{Cat}_{\infty}^{\text{perf}} \rightarrow \text{Cat}_{\infty}^{\text{perf}}$. Then we get the square loop theorem.

Theorem 3.18. *Let $\mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be a small stable idempotent-complete ∞ -category. Then we have a homotopy equivalence*

$$\mathcal{U}_{\text{loc}}(\Theta(\mathcal{C})) \simeq \Omega^2 \mathcal{U}_{\text{loc}}(\mathcal{C}).$$

Besides, Θ is functorial.

Proof. By Theorem 3.9 and Proposition 3.17, we have a homotopy equivalence

$$\mathcal{U}_{\text{loc}}(\Theta(\mathcal{C})) \simeq \text{fib}(\mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{C})) \xrightarrow{\mathcal{U}_{\text{loc}}(\Delta)} \mathcal{U}_{\text{loc}}(\mathbf{BGap}^w(\mathcal{C}))).$$

By Theorem 3.16, we have

$$\text{fib}(\mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{C})) \xrightarrow{\mathcal{U}_{\text{loc}}(\Delta)} \mathcal{U}_{\text{loc}}(\mathbf{BGap}^w(\mathcal{C}))) \simeq \Omega \text{cofib}(\mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{C})) \xrightarrow{\mathcal{U}_{\text{loc}}(\Delta)} \mathcal{U}_{\text{loc}}(\mathbf{BGap}^w(\mathcal{C}))) \simeq \Omega^2 \mathcal{U}_{\text{loc}}(\mathcal{C}).$$

Hence, we have a homotopy equivalence

$$\mathcal{U}_{\text{loc}}(\Theta(\mathcal{C})) \simeq \Omega^2 \mathcal{U}_{\text{loc}}(\mathcal{C}).$$

$\square$

Remark 3.19. [RSW24] gives a functor $\Gamma : \text{Cat}_{\infty}^{\text{perf}} \rightarrow \text{Cat}_{\infty}^{\text{perf}}$ with $\mathcal{U}_{\text{loc}}(\Gamma(\mathcal{C})) \simeq \Omega \mathcal{U}_{\text{loc}}(\mathcal{C})$: $\Gamma(\mathcal{C}) := \text{Cone}(\Delta)$. It also use Proposition 3.7 and Theorem 3.16.

Denote by $\text{Calk}(\mathcal{C}) := \text{Idem}(\text{Ind}(\mathcal{C})^{\omega_1} / \mathcal{C})$ the ω_1 -small Calkin category of $\mathcal{C}$. Then $\mathcal{U}_{\text{loc}}(\text{Calk}(\mathcal{C})) \simeq \Sigma \mathcal{U}_{\text{loc}}(\mathcal{C})$ since $\text{Ind}(\mathcal{C})^{\omega_1}$ admits an Eilenberg swindle.

Corollary 3.20. *Let $\mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be a small stable idempotent-complete ∞ -category. Then*

$$\mathcal{U}_{\text{loc}}(\Theta \circ \text{Calk}(\mathcal{C})) \simeq \Omega \mathcal{U}_{\text{loc}}(\mathcal{C}) \simeq \mathcal{U}_{\text{loc}}(\text{Calk} \circ \Theta(\mathcal{C}))$$

Remark 3.21. [Hen17] defines $\text{Tate}(\mathcal{C})$ with $K(\text{Tate}(\mathcal{C})) \simeq \Sigma K(\mathcal{C})$. So, we can also use Tate here.

Remark 3.22. In fact, we can give a simple proof of Theorem 3.16 now.

A proof of Theorem 3.16. Firstly, observation shows that $\mathbf{BGap}^w(\mathcal{C}) \simeq \text{Gap}^w(\delta_{\mathcal{C}})$. Then we have a diagram of cofiber sequences

$$\begin{array}{ccccc} \mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{C})) & \longrightarrow & \mathcal{U}_{\text{loc}}(\text{Gap}(\mathcal{C})) & \longrightarrow & \mathcal{U}_{\text{loc}}(\mathcal{C}) \\ \downarrow \mathcal{U}_{\text{loc}}(\Delta) & & \downarrow \mathcal{U}_{\text{loc}}(\Delta) & & \downarrow \mathcal{U}_{\text{loc}}(\Delta) \\ \mathcal{U}_{\text{loc}}(\text{Gap}^w(\delta_{\mathcal{C}})) & \longrightarrow & \mathcal{U}_{\text{loc}}(\text{Gap}(\delta_{\mathcal{C}})) & \longrightarrow & \mathcal{U}_{\text{loc}}(\mathcal{C}^2) \end{array}$$

by Proposition 2.13. The middle $\mathcal{U}_{\text{loc}}(\Delta)$ is a homotopy equivalence because the second map and entire map in

$$\mathcal{U}_{\text{loc}}(\text{Gap}(\mathcal{C})) \xrightarrow{\mathcal{U}_{\text{loc}}(\Delta)} \mathcal{U}_{\text{loc}}(\text{Gap}(\mathcal{C}^2, \mathcal{C})) \rightarrow \mathcal{U}_{\text{loc}}(\text{colim}_n \mathcal{C}^n)$$

are homotopy equivalences from Proposition 3.2. Then by Lemma 1.5, there is a cofiber sequence

$$\text{cofib}(\mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{C})) \xrightarrow{\mathcal{U}_{\text{loc}}(\Delta)} \mathcal{U}_{\text{loc}}(\mathbf{BGap}^w(\mathcal{C}))) \rightarrow 0 \rightarrow \text{cofib}(\mathcal{U}_{\text{loc}}(\mathcal{C}) \xrightarrow{\mathcal{U}_{\text{loc}}(\Delta)} \mathcal{U}_{\text{loc}}(\mathcal{C}^2)) \simeq \mathcal{U}_{\text{loc}}(\mathcal{C}).$$

Hence,

$$\text{cofib}(\mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{C})) \xrightarrow{\mathcal{U}_{\text{loc}}(\Delta)} \mathcal{U}_{\text{loc}}(\mathbf{BGap}^w(\mathcal{C}))) \simeq \Omega \mathcal{U}_{\text{loc}}(\mathcal{C}).$$

$\square$

3.5 Remarks on Raptis' dévissage theorem

Ideas of Section 2 and Section 3 are from [Rap18], which gives a dévissage-type theorem for Waldhausen's K-theory [Wal85]. Let's recall some notation from this.

Definition 3.23. Let $\mathcal{C}$ be a Waldhausen category and $\mathcal{A} \subseteq \mathcal{C}$ be a Waldhausen subcategory. We say $(\mathcal{C}, \mathcal{A})$ satisfies the *dévissage condition* if for every morphism $f : X \rightarrow Y$ in $\mathcal{C}$, there is a weak equivalence $g : Y \xrightarrow{\sim} Y'$, $gf : X \rightarrow Y'$ admits a factorization

$$X = X_0 \rightarrowtail X_1 \rightarrowtail \cdots \rightarrowtail X_m \xrightarrow{\sim} Y',$$

where $X_i/X_{i-1} \in \mathcal{A}$ for all $i \geq 1$.

Definition 3.24. Let $\mathcal{C}$ be a Waldhausen category. Define $S_n \mathcal{C}$ be the full subcategory of $\text{Fun}([n], \mathcal{C})$ spanned by filtrations

$$X : 0 = X_0 \rightarrowtail \cdots \rightarrowtail X_n.$$

It is also a Waldhausen category. Let $\mathcal{A} \subseteq \mathcal{C}$ be a Waldhausen subcategory. Define $S_n \mathcal{C}_{\mathcal{A}}$ by the subcategory of $S_n \mathcal{C}$:

- Objects are X such that X_i/X_{i-1} are objects in $\mathcal{A}$ for all $i \geq 1$.
- Morphisms are $X \rightarrow Y$ such that $X_i/X_{i-1} \rightarrow Y_i/Y_{i-1}$ are morphisms in $\mathcal{A}$ for all $i \geq 1$.

The map $[n+1] \rightarrow [n], n+1 \mapsto n$ induces $S_n \mathcal{C} \rightarrow S_{n+1} \mathcal{C}$ and $S_n \mathcal{C}_{\mathcal{A}} \rightarrow S_{n+1} \mathcal{C}_{\mathcal{A}}$. Thus, we can define

$$S_{\infty} \mathcal{C} := \text{colim}_n S_n \mathcal{C}, S_{\infty} \mathcal{C}_{\mathcal{A}} := \text{colim}_n S_n \mathcal{C}_{\mathcal{A}}.$$

Definition 3.25. For $n \in \mathbb{N}$, define $S_n^w \mathcal{C}_{\mathcal{A}}$ be the full subcategory of $S_n \mathcal{C}_{\mathcal{A}}$ spanned by objects X such that $0 \rightarrow X_n$ is weak equivalence. Define $S_{\infty}^w \mathcal{C}_{\mathcal{A}} := \text{colim}_n S_n^w \mathcal{C}_{\mathcal{A}}$.

Then we have a table of correspondence between section 2,3 in this article and [Rap18].

Exact functor $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C}$ between idempotent-complete stable ∞ -categories satisfies the dévissage condition		A pair of a good Waldhausen category $\mathcal{C}$ and good Waldhausen subcategory $\mathcal{A}$ satisfies the dévissage condition	
Result	Label	Result	label
$K(\text{Gap}(\mathcal{F})) \simeq \text{colim}_n K(\mathcal{A})^n$	Proposition 3.2	$K(S_{\infty} \mathcal{C}_{\mathcal{A}}) \simeq \text{colim}_n K(\mathcal{A})^n$	[Rap18, Proposition 3.2]
$K(\text{Gap}^w(\mathcal{F})) \rightarrow K(\text{Gap}(\mathcal{F})) \rightarrow K(\mathcal{C})$	Proposition 2.13	$K(S_{\infty}^w \mathcal{C}_{\mathcal{A}}) \rightarrow K(S_{\infty} \mathcal{C}_{\mathcal{A}}) \rightarrow K(\mathcal{C})$	[Rap18, Proposition 3.7, 5.5]
$K(Q(\mathcal{F})) \simeq \text{fib } K(\mathcal{F})$	Theorem 3.9	Raptis' dévissage theorem	[Rap18, Theorem 5.7]

4 FILLABILITY

4.1 1-fillable functors

For a small stable ∞ -categories $\mathcal{C}$, its connective K_0^{cn} -group is the Grothendieck group of $\text{Ho}(\mathcal{C})$:

$$K_0^{\text{cn}}(\mathcal{C}) = \frac{\text{Free abelian group on isomorphism classes in } \text{Ho}(\mathcal{C})}{\langle [X] - [Y] + [Z] : \text{cofiber sequence } X \rightarrow Y \rightarrow Z \rangle}$$

Its non-connective K_0 -group is $K_0(\mathcal{C}) = K^{\text{cn}}(\text{Idem}(\mathcal{C}))$.

Lemma 4.1. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ satisfies the dévissage condition,

$$K_0(\mathcal{F}) : K_0(\mathcal{A}) \rightarrow K_0(\mathcal{C})$$

is an epimorphism.

Proof. Just from the definition of dévissage condition and the K_0 groups. □

Denote by $\text{Gap}^\vee(\mathcal{F})$ the pullback

$$\begin{array}{ccc} \text{Gap}^\vee(\mathcal{F}) & \longrightarrow & \text{Gap}(\mathcal{F}) \\ \downarrow & & \downarrow \text{ev}_{\mathcal{F}} \\ \text{Gap}(\mathcal{F}) & \xrightarrow{\text{ev}_{\mathcal{F}}} & \mathcal{C} \end{array}$$

Hence, it is stable. Denote $\text{Gap}^\vee(\mathcal{C}) := \text{Gap}^\vee(\text{id}_{\mathcal{C}})$.

Define an exact functor $\text{Gap}_{\mathcal{F}} : \text{Gap}(\mathcal{A}) \rightarrow \text{Gap}(\mathcal{F})$ by $\text{Gap}(\mathcal{A}) \rightarrow \text{Gap}(\mathcal{C}), X_{\bullet, \bullet} \mapsto \mathcal{F}(X_{\bullet, \bullet})$ and $\text{gr} : \text{Gap}(\mathcal{A}) \rightarrow \text{colim}_n \mathcal{A}^n$ in last section. Then it induces an exact functor $\text{Gap}_{\mathcal{F}}^\vee : \text{Gap}^\vee(\mathcal{A}) \rightarrow \text{Gap}^\vee(\mathcal{F})$.

Definition 4.2. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. We call $\mathcal{F}$ 1-*fillable* if $\text{Gap}_{\mathcal{F}}^\vee$ satisfies the dévissage condition.

Define exact functors

$$\tau_{\perp} : \text{Gap}^w(\mathcal{F}) \rightarrow \text{Gap}^\vee(\mathcal{F}), (A_{\bullet}, X_{\bullet, \bullet}) \mapsto ((A_{\bullet}, X_{\bullet, \bullet}), (0_{\bullet}, 0_{\bullet, \bullet}))$$

and

$$p_{\lrcorner} : \text{Gap}^\vee(\mathcal{F}) \rightarrow \text{Gap}(\mathcal{F}), ((A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet})) \mapsto (B_{\bullet}, Y_{\bullet, \bullet}).$$

Proposition 4.3.

$$\text{Gap}^w(\mathcal{F}) \xrightarrow{\tau_{\perp}} \text{Gap}^\vee(\mathcal{F}) \xrightarrow{p_{\lrcorner}} \text{Gap}(\mathcal{F})$$

is an exact sequence. Besides, the cofiber sequence

$$K(\text{Gap}^w(\mathcal{F})) \xrightarrow{K(\tau_{\perp})} K(\text{Gap}^\vee(\mathcal{F})) \xrightarrow{K(p_{\lrcorner})} K(\text{Gap}(\mathcal{F}))$$

split.

Proof. By [BGT13, Proposition 5.15], we only need to prove the exact sequence in the sense of homotopy categories.

For two objects $(A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet}) \in \text{Gap}^w(\mathcal{F})$, we have

$$\begin{aligned} \text{Map}_{\text{Gap}^\vee(\mathcal{F})}(\tau_{\perp}(A_{\bullet}, X_{\bullet, \bullet}), \tau_{\perp}(B_{\bullet}, Y_{\bullet, \bullet})) &\simeq \text{Map}_{\text{Gap}(\mathcal{F})}((A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet})) \times_{\text{Map}_{\mathcal{C}}(0, 0)} \text{Map}_{\text{Gap}(\mathcal{F})}(0, 0) \\ &\simeq \text{Map}_{\text{Gap}(\mathcal{F})}((A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet})) \\ &\simeq \text{Map}_{\text{Gap}^w(\mathcal{F})}((A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet})) \end{aligned}$$

Hence, $\tau_{\perp}$ is fully faithful. Since $p_{\lrcorner}$ admits a section

$$\Delta : \text{Gap}(\mathcal{F}) \rightarrow \text{Gap}^\vee(\mathcal{F}), (A_{\bullet}, X_{\bullet, \bullet}) \mapsto ((A_{\bullet}, X_{\bullet, \bullet}), (A_{\bullet}, X_{\bullet, \bullet})),$$

$p_{\lrcorner}$ is essentially surjective on both objects and morphisms. Let $((A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet})) \in \text{Gap}^\vee(\mathcal{F})$, there are two morphisms $(A_{\bullet}, X_{\bullet, \bullet}) \rightarrow (C_{\bullet}, Z_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet}) \rightarrow (C_{\bullet}, Z_{\bullet, \bullet})$ which are stable at identities from Proposition 2.13. Hence, there are two morphisms

$$((A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet})) \rightarrow ((C_{\bullet}, Z_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet})), ((B_{\bullet}, Y_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet})) \rightarrow ((C_{\bullet}, Z_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet}))$$

whose cofibers are in $\tau_{\perp}(\text{Gap}^w(\mathcal{F}))$. Hence,

$$\text{Gap}^\vee(\mathcal{F}) / \text{Gap}^w(\mathcal{F}) \rightarrow \text{Gap}(\mathcal{F})$$

is injective on objects. Similarly, from Proposition 2.13, it is also injective on morphisms. Hence, it is an equivalence in the sense of homotopy categories.

Then we have a cofiber sequence

$$K(\text{Gap}^w(\mathcal{F})) \xrightarrow{K(\tau_{\perp})} K(\text{Gap}^\vee(\mathcal{F})) \xrightarrow{K(p_{\lrcorner})} K(\text{Gap}(\mathcal{F})).$$

Since $K(p_{\lrcorner})$ admits a section $K(\Delta)$, it is split. □

Theorem 4.4. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be a 1-fillable exact functor between small stable idempotent-complete ∞ -categories. Then $K_0(Q(\mathcal{F})) = 0$. Furthermore, if $\mathcal{F}$ satisfies the dévissage condition, $K_0(\mathcal{F}) : K_0(\mathcal{A}) \xrightarrow{\cong} K_0(\mathcal{C})$ is an isomorphism.

Proof. $\text{Gap}_{\mathcal{F}}$ restricts to a functor $\text{Gap}_{\mathcal{F}}^w : \text{Gap}^w(\mathcal{A}) \rightarrow \text{Gap}^w(\mathcal{F})$. Consider the diagram

$$\begin{array}{ccccc} \text{Gap}^w(\mathcal{A}) & \longrightarrow & \text{Gap}^v(\mathcal{A}) & \longrightarrow & \text{Gap}(\mathcal{A}) \\ \downarrow \text{Gap}_{\mathcal{F}}^w & & \downarrow \text{Gap}_{\mathcal{F}}^v & & \downarrow \text{Gap}_{\mathcal{F}} \\ \text{Gap}^w(\mathcal{F}) & \longrightarrow & \text{Gap}^v(\mathcal{F}) & \longrightarrow & \text{Gap}(\mathcal{F}) \end{array}$$

By Proposition 4.3, $K_0(\text{Gap}_{\mathcal{F}}^v)$ is just the direct sum of $K_0(\text{Gap}_{\mathcal{F}}^w)$ and $K_0(\text{Gap}_{\mathcal{F}})$. By Lemma 4.1, $K_0(\text{Gap}_{\mathcal{F}}^w)$ is an epimorphism. Consider the diagram of two split cofiber sequence

$$\begin{array}{ccccc} K(\text{colim}_n \mathcal{A}^n) & \xrightarrow{K(\tau_{\text{id}_{\mathcal{A}}})} & K(\text{Gap}^w(\mathcal{A})) & \longrightarrow & K(Q(\text{id}_{\mathcal{A}})) \\ \downarrow \text{id} & & \downarrow K(\text{Gap}_{\mathcal{F}}^w) & & \downarrow \\ K(\text{colim}_n \mathcal{A}^n) & \xrightarrow{K(\tau_{\mathcal{F}})} & K(\text{Gap}^w(\mathcal{F})) & \longrightarrow & K(Q(\mathcal{F})) \end{array}$$

Then $K_0(Q(\text{id}_{\mathcal{A}})) \rightarrow K_0(Q(\mathcal{F}))$ is an epimorphism. Since $K(Q(\text{id}_{\mathcal{A}})) \simeq \text{fib}(K(\text{id}_{\mathcal{A}})) = 0$ by Theorem 3.9, $K_0(Q(\mathcal{F})) = 0$.

If $\mathcal{F}$ satisfies the dévissage condition, by Lemma 4.1, $K_0(\mathcal{F})$ is an epimorphism. By Theorem 3.9, there is a cofiber sequence

$$K(Q(\mathcal{F})) \rightarrow K(\mathcal{A}) \xrightarrow{K(\mathcal{F})} K(\mathcal{C}).$$

Since $K_0(Q(\mathcal{F})) = 0$, $K_0(\mathcal{F})$ is a monomorphism. Hence, $K_0(\mathcal{F})$ is an isomorphism. $\square$

Example 4.5. $\text{id}_{\mathcal{C}}$ is always 1-fillable: $\text{Gap}^v(\text{id}_{\mathcal{C}}) \simeq \text{Gap}^v(\mathcal{C})$ and $\text{Gap}_{\text{id}_{\mathcal{C}}}^v \simeq \text{id}_{\text{Gap}^v(\mathcal{C})}$ is also the identity map.

$\text{Gap}([n], \text{Gap}([n], \mathcal{C}))$ is a full subcategory of $\text{Fun}(N(\text{Ar}[n] \times \text{Ar}[n]), \mathcal{C})$, whose objects are denoted by $X_{\bullet, \bullet}^{i, j}$, where for every i, j , $X_{\bullet, \bullet}^{i, j}, X_{\bullet, \bullet}^{i, j}$ are objects in $\text{Gap}(\mathcal{C})$. By [BGT13, Lemma 7.3], an object $X_{\bullet, \bullet}^{i, j}$ is decided by $X_{0, \bullet}^{0, j}$, the 0-the grid. Hence, an object $X_{\bullet, \bullet}^{i, j}$ in $\text{Gap}(\text{Gap}(\mathcal{C}))$ has two sides and one vertex at infinity, which is an object in $\text{Gap}^v(\mathcal{C})$. This actually means that, whether we look at it from rows or columns, it is an object within $\text{Gap}(\text{Gap}(\mathcal{C}))$. From this, we stipulate the evaluation functor is in the sense of columns:

$$\text{ev}_{\text{id}_{\text{Gap}(\mathcal{C})}} : \text{Gap}(\text{Gap}(\mathcal{C})) \rightarrow \text{Gap}(\mathcal{C}), X_{\bullet, \bullet}^{i, j} \mapsto \left(0 \rightarrow \text{colim}_n X_{0, n}^{0, 1} \rightarrow \cdots \rightarrow \text{colim}_n X_{0, n}^{0, m} \rightarrow \cdots \right)$$

And, we have a "transpose" functor, which swaps rows and columns:

$$T_{\mathcal{C}} : \text{Gap}(\text{Gap}(\mathcal{C})) \rightarrow \text{Gap}(\text{Gap}(\mathcal{C})), T_{\mathcal{C}}(X_{\bullet, \bullet}^{i, j})_{i, j}^{k, l} = X_{k, l}^{i, j},$$

which induces a transpose evaluation: $\text{ev}_{\text{id}_{\text{Gap}(\mathcal{C})}}^T := \text{ev}_{\text{id}_{\text{Gap}(\mathcal{C})}} \circ T_{\mathcal{C}}$.

Objects $\text{Gap}(\text{Gap}_{\mathcal{F}})$ are three tuples $(A_{\bullet, \bullet}^{\bullet}, X_{\bullet, \bullet}^{i, j}, A_{\bullet, \bullet}^{\bullet})$, where $A_{\bullet, \bullet}^{\bullet}, A_{\bullet, \bullet}^{\bullet} \in \text{colim}_n (\text{Gap}(\mathcal{A}))^n$ and $X_{\bullet, \bullet}^{i, j} \in \text{Gap}(\text{Gap}(\mathcal{C}))$ such that for each i, j , $(A_{\bullet, \bullet}^{\bullet}, X_{\bullet, \bullet}^{i, j}), (A_{\bullet, \bullet}^{\bullet}, X_{\bullet, \bullet}^{i, j}) \in \text{Gap}(\mathcal{F})$. We also stipulate the evaluation, transpose and the transpose evaluation functors as follows:

$$\begin{aligned} \text{ev}_{\text{Gap}_{\mathcal{F}}} : \text{Gap}(\text{Gap}_{\mathcal{F}}) &\rightarrow \text{Gap}(\mathcal{F}), (A_{\bullet, \bullet}^{\bullet}, X_{\bullet, \bullet}^{i, j}, A_{\bullet, \bullet}^{\bullet}) \mapsto \left(\text{colim}_n A_{0, n}^{\bullet}, \text{ev}_{\text{id}_{\text{Gap}(\mathcal{C})}}(X_{\bullet, \bullet}^{i, j}) \right) \\ T_{\mathcal{F}} : \text{Gap}(\text{Gap}_{\mathcal{F}}) &\rightarrow \text{Gap}(\text{Gap}_{\mathcal{F}}), (A_{\bullet, \bullet}^{\bullet}, X_{\bullet, \bullet}^{i, j}, A_{\bullet, \bullet}^{\bullet}) \mapsto (A_{\bullet, \bullet}^{\bullet}, T_{\mathcal{C}}(X_{\bullet, \bullet}^{i, j}), A_{\bullet, \bullet}^{\bullet}) \\ \text{ev}_{\text{Gap}_{\mathcal{F}}}^T &:= \text{ev}_{\text{Gap}_{\mathcal{F}}} \circ T_{\mathcal{F}} \end{aligned}$$

Then we can define a "forgetful" functor $U_{\mathcal{F}} : \text{Gap}(\text{Gap}_{\mathcal{F}}) \rightarrow \text{Gap}^v(\mathcal{F})$ by $(\text{ev}_{\text{Gap}_{\mathcal{F}}}^T, \text{ev}_{\text{Gap}_{\mathcal{F}}})$, the composition

$$\text{Gap}(\text{Gap}_{\mathcal{F}}) \rightarrow \text{Gap}(\text{Gap}_{\mathcal{F}}^v) \xrightarrow{\text{ev}_{\text{Gap}_{\mathcal{F}}^v}} \text{Gap}^v(\mathcal{F}).$$

Proposition 4.6. The following are equivalent:

- (1) $\mathcal{F}$ is 1-fillable.
- (2) $U_{\mathcal{F}}$ is essentially surjective.

Proof. By Corollary 2.14, $\mathcal{F}$ is 1-fillable if and only if $\text{ev}_{\text{Gap}_{\mathcal{F}}^{\vee}}$ is essentially surjective.

$U_{\mathcal{F}}$ is the composition

$$\text{Gap}(\text{Gap}_{\mathcal{F}}) \rightarrow \text{Gap}(\text{Gap}_{\mathcal{F}}^{\vee}) \simeq \text{Gap}^{\vee}(\text{Gap}_{\mathcal{F}}) \xrightarrow{\text{ev}_{\text{Gap}_{\mathcal{F}}^{\vee}}} \text{Gap}^{\vee}(\mathcal{F}),$$

where the first map $\text{Gap}(\text{Gap}_{\mathcal{F}}) \rightarrow \text{Gap}^{\vee}(\text{Gap}_{\mathcal{F}})$ is defined by $(T_{\mathcal{F}}, \text{id})$.

Hence, (2) $\Rightarrow$ (1) is obvious. Now consider (1) $\Rightarrow$ (2).

For $((A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet})) \in \text{Gap}^{\vee}(\mathcal{F})$, assume there is an object

$$((A_{\bullet, \bullet}^{\bullet}, X_{\bullet, \bullet}^{\bullet}, A_{\bullet, \bullet}^{\bullet}), (B_{\bullet, \bullet}^{\bullet}, Y_{\bullet, \bullet}^{\bullet}, B_{\bullet, \bullet}^{\bullet})) \in \text{Gap}([n], \text{Gap}_{\mathcal{F}}^{\vee})$$

such that $(A_{0, n}^{\bullet}, X_{0, n}^{\bullet}) = (A_{\bullet}, X_{\bullet, \bullet}), (B_{0, n}^{\bullet}, Y_{0, n}^{\bullet}) = (B_{\bullet}, Y_{\bullet, \bullet})$. Also, we can assume $X_{0, i}^{\bullet}, Y_{0, i}^{\bullet} \in \text{Gap}([m], \mathcal{F})$ for all $1 \leq i \leq n$. It means $X_{0, i}^{0, m} = Y_{0, i}^{0, m}$ for all $1 \leq i \leq n$. Then define $Z_{\bullet, \bullet}^{\bullet} \in \text{Gap}(\text{Gap}(\mathcal{C}))$ by

$$\begin{array}{ccccccccccc} 0 & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 \\ \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & & & \downarrow \\ 0 & \longrightarrow & \tilde{X}_{0,1}^{0,1} & \longrightarrow & \cdots & \longrightarrow & \tilde{X}_{0,1}^{0,m} = Y_{0,1}^{0,1} & \longrightarrow & Y_{0,2}^{0,1} & \longrightarrow & \cdots & \longrightarrow & Y_{0,n}^{0,1} \\ \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & & & \downarrow \\ \vdots & & \vdots & & \ddots & & \vdots & & \vdots & & \ddots & & \vdots \\ \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & & & \downarrow \\ 0 & \longrightarrow & \tilde{X}_{0,m}^{0,1} = X_{0,1}^{0,1} & \longrightarrow & \cdots & \longrightarrow & \tilde{X}_{0,m}^{0,m} = X_{0,1}^{0,m} & \longrightarrow & X_{0,2}^{0,m} & \longrightarrow & \cdots & \longrightarrow & X_{0,n}^{0,m} \\ \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & & & \downarrow \\ 0 & \longrightarrow & X_{0,2}^{0,1} & \longrightarrow & \cdots & \longrightarrow & X_{0,2}^{0,m} & \longrightarrow & X_{0,2}^{0,m} & \longrightarrow & \cdots & \longrightarrow & X_{0,n}^{0,m} \\ \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & & & \downarrow \\ \vdots & & \vdots & & \ddots & & \vdots & & \vdots & & \ddots & & \vdots \\ \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & & & \downarrow \\ 0 & \longrightarrow & X_{0,n}^{0,1} & \longrightarrow & \cdots & \longrightarrow & X_{0,n}^{0,m} & \longrightarrow & X_{0,n}^{0,m} & \longrightarrow & \cdots & \longrightarrow & X_{0,n}^{0,m} \end{array}$$

where for $1 \leq i, j \leq m$, $\tilde{X}_{0,j}^{0,i} := X_{0,1}^{0,i} \times_{X_{0,1}^{0,m}} Y_{0,1}^{0,j}$. Besides, define $C_{\bullet, \bullet}^{\bullet}, C_{\bullet, \bullet}^{\bullet}$ by:

$$C_{0,j}^i = \begin{cases} A_{0,1}^i, & 1 \leq i, j \leq m \\ B_{0,j-m+1}^i, & 1 \leq i \leq m, m < j \leq m+n-1 \\ A_j^{0,i-m+1}, & 1 \leq j \leq m, m < i \leq m+n-1 \\ A_{0,i-m+1}^{j-m+1}, & m < i, j \leq m+n-1 \\ 0, & \text{otherwise} \end{cases}$$

and

$$C_i^{0,j} = \begin{cases} B_i^{0,1}, & 1 \leq i, j \leq m \\ A_i^{0,j-m+1}, & 1 \leq i \leq m, m < j \leq m+n-1 \\ B_{0,i-m+1}^j, & 1 \leq j \leq m, m < i \leq m+n-1 \\ B_{0,i-m+1}^{j-m+1}, & m < i, j \leq m+n-1 \\ 0, & \text{otherwise} \end{cases}$$

The image of $(C_{\bullet, \bullet}^{\bullet}, X_{\bullet, \bullet}^{\bullet}, C_{\bullet, \bullet}^{\bullet})$ under $U_{\mathcal{F}}$ is just $((A_{\bullet}, X_{\bullet, \bullet}), (B_{\bullet}, Y_{\bullet, \bullet}))$. □

Remark 4.7. The essentially surjection of $U_{\mathcal{F}}$ just means that given two infinity sides with the same infinity vertex in $\text{Gap}^{\vee}(\mathcal{F})$, we can "fill" it into a whole grid in $\text{Gap}(\text{Gap}_{\mathcal{F}})$. This is the origin of the name "fillability".

4.2 Higher fillability for higher K -groups

Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. Define exact functors $\mathcal{F}_n : \mathcal{A}_n \rightarrow \mathcal{C}_n$ by

- $\mathcal{C}_1 := \mathcal{C}, \mathcal{A}_1 := \mathcal{A}, \mathcal{F}_1 := \mathcal{F}$.
- For $n \geq 2$, $\mathcal{C}_n := \text{Gap}^{\vee}(\mathcal{F}_{n-1}), \mathcal{A}_n := \text{Gap}^{\vee}(\mathcal{A}_{n-1})$ and $\mathcal{F}_n := \text{Gap}_{\mathcal{F}_{n-1}}^{\vee}$.

Definition 4.8. $\mathcal{F}$ is called *n-fillable* if for every $1 \leq i \leq n$, $\mathcal{F}_i$ is 1-fillable. $\mathcal{F}$ is called *fillable* for every $i \geq 1$, $\mathcal{F}_i$ is 1-fillable.

Proposition 4.9. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories.

- (1) $\mathcal{F}$ is *n-fillable* if and only if $\mathcal{F}_i$ satisfies the dévissage condition for all $2 \leq i \leq n+1$.
- (2) $\mathcal{F}$ is fillable if and only if $\mathcal{F}_n$ is fillable for all $n \geq 1$.

Proof. Just definition. □

Lemma 4.10. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ is *n-fillable*, $\mathcal{U}_{\text{loc}}(\mathcal{Q}(\mathcal{F}_{k+1})) \simeq \Omega \mathcal{U}_{\text{loc}}(\mathcal{Q}(\mathcal{F}_k))$ for all $1 \leq k \leq n$. As a consequence, $\mathcal{U}_{\text{loc}}(\mathcal{Q}(\mathcal{F}_{k+1})) \simeq \Omega^k \mathcal{U}_{\text{loc}}(\mathcal{Q}(\mathcal{F}))$.

Proof. From Proposition 4.9, $\mathcal{F}_{k+1}$ satisfies the dévissage condition for all $1 \leq k \leq n$. By Theorem 3.9, there is a cofiber sequence

$$\mathcal{U}_{\text{loc}}(\mathcal{Q}(\mathcal{F}_{k+1})) \simeq \text{fib}(\mathcal{U}_{\text{loc}}(\mathcal{F}_{k+1})).$$

By Proposition 4.3,

$$\mathcal{U}_{\text{loc}}(\mathcal{A}_{k+1}) \simeq \mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{A}_k)) \oplus \mathcal{U}_{\text{loc}}(\text{Gap}(\mathcal{A}_k)), \mathcal{U}_{\text{loc}}(\mathcal{C}_{k+1}) \simeq \mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{F}_k)) \oplus \mathcal{U}_{\text{loc}}(\text{Gap}(\mathcal{F}_k)).$$

Hence,

$$\begin{aligned} & \text{fib}(\mathcal{U}_{\text{loc}}(\mathcal{F}_{k+1})) \\ & \simeq \text{fib}(\mathcal{U}_{\text{loc}}(\text{Gap}_{\mathcal{F}_k}^w)) \oplus \text{fib}(\mathcal{U}_{\text{loc}}(\text{Gap}_{\mathcal{F}_k})). \end{aligned}$$

The first part $\text{fib}(\mathcal{U}_{\text{loc}}(\text{Gap}_{\mathcal{F}_k}^w))$ is just $\Omega \mathcal{U}_{\text{loc}}(\mathcal{Q}(\mathcal{F}_k))$. And since $\mathcal{U}_{\text{loc}}(\text{Gap}(\mathcal{F}_k))$ is an equivalence from Proposition 3.2, $\text{fib}(\mathcal{U}_{\text{loc}}(\text{Gap}_{\mathcal{F}_k})) \simeq 0$. Thus,

$$\mathcal{U}_{\text{loc}}(\mathcal{Q}(\mathcal{F}_{k+1})) \simeq \Omega \mathcal{U}_{\text{loc}}(\mathcal{Q}(\mathcal{F}_k)).$$

□

Theorem 4.11. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ is *n-fillable*, $K_i(\mathcal{Q}(\mathcal{F})) = 0$ for all $0 \leq i \leq n-1$.

Proof. By Theorem 4.4, $K_0(\mathcal{Q}(\mathcal{F}_i)) = 0$ for all $1 \leq i \leq n$. By Lemma 4.10, $K_i(\mathcal{Q}(\mathcal{F})) = 0$ for all $0 \leq i \leq n-1$. □

Theorem 4.12. (Theorem A) Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be a *n-fillable* exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ satisfies the weak dévissage condition (resp. dévissage condition), it induces isomorphisms $K_i(\mathcal{A}) \xrightarrow{\cong} K_i(\mathcal{C})$ for all $1 \leq i \leq n-1$, an epimorphism $K_n(\mathcal{A}) \twoheadrightarrow K_n(\mathcal{C})$ and a monomorphism (resp. an isomorphism) $K_0(\mathcal{A}) \hookrightarrow K_0(\mathcal{C})$.

Proof. Just from Theorem 4.11 and Theorem 3.9. □

Corollary 4.13. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be a fillable exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ satisfies the weak dévissage condition (resp. dévissage condition), it induces isomorphisms $K_n(\mathcal{A}) \xrightarrow{\cong} K_n(\mathcal{C})$ for all $n \geq 1$, and a monomorphism (resp. an isomorphism) $K_0(\mathcal{A}) \hookrightarrow K_0(\mathcal{C})$.

Definition 4.14. Let $\mathcal{C}$ be a small stable ∞ -category. $\mathcal{C}$ is called *n-fillable* if the delta map $\delta_{\mathcal{C}} : \mathcal{C} \rightarrow \mathcal{C}^2$ is *n-fillable*. $\mathcal{C}$ is called *fillable* if $\delta_{\mathcal{C}}$ is fillable.

Theorem 4.15. (Theorem B) Let $\mathcal{C}$ be a small idempotent-complete *n-fillable* ∞ -category. Then $K_i(\mathcal{C}) = 0$ for all $1 \leq i \leq n$.

Proof. By Theorem 4.12 and Lemma 3.13, $K_i(\mathcal{C}) \rightarrow K_i(\mathcal{C}^2)$ are surjective for all $1 \leq i \leq n$, which forces them are trivial. $\square$

Corollary 4.16. Let $\mathcal{C}$ be a small idempotent-complete fillable ∞ -category. Then $K_n(\mathcal{C}) = 0$ for all $n \geq 1$.

4.3 Strongly 1-fillability

Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories.

Lemma 4.17. Let J be a finite ∞ -category. $\text{Fun}(J, -) : \text{Cat}_\infty^{\text{ex}} \rightarrow \text{Cat}_\infty^{\text{ex}}$ satisfies:

- (1) It preserves all filtered colimits.
- (2) If $\mathcal{C}$ is a full subcategory of $\mathcal{D}$, then $\text{Fun}(J, \mathcal{C})$ is a full subcategory of $\text{Fun}(J, \mathcal{D})$. Besides, $\text{Fun}(J, \mathcal{D}/\mathcal{C}) \simeq \text{Fun}(J, \mathcal{D})/\text{Fun}(J, \mathcal{C})$.

Proof.

- (1) It is because J is a compact object in Cat_∞ and the inclusion $\text{Cat}_\infty^{\text{ex}} \rightarrow \text{Cat}_\infty$ preserves filtered colimits [Lur17, Proposition 1.1.4.6].
- (2) Mapping spaces in a functor category are computed as limits of objectwise mapping spaces:

$$\text{Map}_{\text{Fun}(J, \mathcal{D})}(F, G) \simeq \lim_{j \in J} \text{Map}_{\mathcal{D}}(F(j), G(j)).$$

If F, G land in $\mathcal{C}$ and $\mathcal{C} \subseteq \mathcal{D}$ is fully faithful, then

$$\text{Map}_{\mathcal{C}}(F(j), G(j)) \xrightarrow{\cong} \text{Map}_{\mathcal{D}}(F(j), G(j))$$

is an equivalence for each j , hence taking the finite limit over $j \in J$ gives

$$\text{Map}_{\text{Fun}(J, \mathcal{C})}(F, G) \simeq \text{Map}_{\text{Fun}(J, \mathcal{D})}(F, G).$$

So the inclusion is fully faithful.

The map $\mathcal{D} \rightarrow \mathcal{D}/\mathcal{C}$ induces a map

$$\Phi : \text{Fun}(J, \mathcal{D}) \rightarrow \text{Fun}(J, \mathcal{D}/\mathcal{C}).$$

One can check that $\Phi(\text{Fun}(J, \mathcal{C})) = \{0\}$. Hence, it induces a functor

$$\tilde{\Phi} : \text{Fun}(J, \mathcal{D})/\text{Fun}(J, \mathcal{C}) \rightarrow \text{Fun}(J, \mathcal{D}/\mathcal{C}).$$

It is essentially surjective since Φ is. Denote

$$W := \{\eta : F \rightarrow F' \mid \text{the cofiber } \text{cofib}(\eta) \in \text{Fun}(J, \mathcal{C})\}$$

For $F, G \in \text{Fun}(J, \mathcal{D})$,

$$\begin{aligned} \text{Map}_{\text{Fun}(J, \mathcal{D})/\text{Fun}(J, \mathcal{C})}(F, G) &\simeq \text{colim}_{\left(F' \xrightarrow{\eta} F\right) \in W/F} \left(\lim_{j \in J} \text{Map}_{\mathcal{D}}(F'(j), G(j)) \right) \\ &\simeq \lim_{j \in J} \left(\text{colim}_{\left(F' \xrightarrow{\eta} F\right) \in W/F} \text{Map}_{\mathcal{D}}(F'(j), G(j)) \right) \\ &\simeq \lim_{j \in J} \text{Map}_{\mathcal{D}/\mathcal{C}}(\tilde{\Phi}(F)(j), \tilde{\Phi}(G)(j)) \\ &\simeq \text{Map}_{\text{Fun}(J, \mathcal{D}/\mathcal{C})}(\tilde{\Phi}(F), \tilde{\Phi}(G)) \end{aligned}$$

It means $\tilde{\Phi}$ is fully faithful; hence, it is an equivalence. Here, we use that finite limits commute with filtered colimits in the ∞ -category of Kan complexes. $\square$

Lemma 4.18. *If $\mathcal{C}$ is a full subcategory of $\mathcal{D}$, then $\text{Gap}^w(\mathcal{C})$ is a full subcategory of $\text{Gap}^w(\mathcal{D})$ and $\text{Gap}^w(\mathcal{D})/\text{Gap}^w(\mathcal{C}) \rightarrow \text{Gap}^w(\mathcal{D}/\mathcal{C})$ is an equivalence.*

Proof. Notice that $\text{Gap}^w \simeq \text{colim}_n \text{Fun}(\Delta^{n-2}, -)$. This is from Lemma 4.17 naturally. $\square$

Lemma 4.19. $\text{BGap}^w : \text{Cat}_{\infty}^{\text{ex}} \rightarrow \text{Cat}_{\infty}^{\text{ex}}, \mathcal{C} \mapsto \text{BGap}^w(\mathcal{C})$ is left exact.

Proof. Since $\text{BGap}^w(*) \simeq *$, from Lemma 1.12, we only need to verify it preserves pullbacks. Notice that $\text{Gap}^w \simeq \text{colim}_n \text{Fun}(\Delta^{n-2}, -)$ is left exact. For a pullback $\mathcal{A} \simeq \mathcal{B} \times_{\mathcal{D}} \mathcal{C}$ in $\text{Cat}_{\infty}^{\text{ex}}$, we have

$$\begin{aligned} \text{BGap}^w(\mathcal{A}) &\simeq \text{Gap}^w(\mathcal{A}) \times_{\text{colim}_n \mathcal{A}^n} \text{Gap}^w(\mathcal{A}) \\ &\simeq \text{Gap}^w(\mathcal{B} \times_{\mathcal{D}} \mathcal{C}) \times_{\text{colim}_n (\mathcal{B} \times_{\mathcal{D}} \mathcal{C})^n} \text{Gap}^w(\mathcal{B} \times_{\mathcal{D}} \mathcal{C}) \\ &\simeq \left(\text{Gap}^w(\mathcal{B}) \times_{\text{Gap}^w(\mathcal{D})} \text{Gap}^w(\mathcal{C}) \right) \times_{\text{colim}_n (\mathcal{B} \times_{\mathcal{D}} \mathcal{C})^n} \left(\text{Gap}^w(\mathcal{B}) \times_{\text{Gap}^w(\mathcal{D})} \text{Gap}^w(\mathcal{C}) \right) \\ &\simeq \text{BGap}^w(\mathcal{B}) \times_{\text{BGap}^w(\mathcal{D})} \text{BGap}^w(\mathcal{C}) \end{aligned}$$

$\square$

Lemma 4.20. *Let J be a finite ∞ -category. Then $\text{Fun}(J, -) \circ \text{BGap}^w \simeq \text{BGap}^w \circ \text{Fun}(J, -)$.*

Proof. Notice that $\text{Gap}^w([n], -) \simeq \text{Fun}(\Delta^{n-2}, -)$. We have $\text{Fun}(J, -) \circ \text{Gap}^w([n], -) \simeq \text{Gap}^w([n], -) \circ \text{Fun}(J, -)$. Since $\text{Fun}(J, -)$ is left exact, we have

$$\begin{aligned} \text{Fun}(J, \text{BGap}^w([n], \mathcal{C})) &\simeq \text{Fun}(J, \text{Gap}^w([n], \mathcal{C})) \times_{\text{Fun}(J, \mathcal{C}^n)} \text{Fun}(J, \text{Gap}^w([n], \mathcal{C})) \\ &\simeq \text{Gap}^w([n], \text{Fun}(J, \mathcal{C})) \times_{\text{Fun}(J, \mathcal{C})^n} \text{Gap}^w([n], \text{Fun}(J, \mathcal{C})) \\ &\simeq \text{BGap}^w([n], \text{Fun}(J, \mathcal{C})) \end{aligned}$$

Hence, $\text{Fun}(J, -) \circ \text{BGap}^w([n], -) \simeq \text{BGap}^w([n], -) \circ \text{Fun}(J, -)$. Since $\text{Fun}(J, -)$ preserves filtered colimits by Lemma 4.17, we have $\text{Fun}(J, -) \circ \text{BGap}^w \simeq \text{BGap}^w \circ \text{Fun}(J, -)$. $\square$

Lemma 4.21. *If $\mathcal{C}$ is a full subcategory of $\mathcal{D}$, then $\text{BGap}^w(\mathcal{C})$ is a full subcategory of $\text{BGap}^w(\mathcal{D})$ and $\text{BGap}^w(\mathcal{D})/\text{BGap}^w(\mathcal{C}) \rightarrow \text{BGap}^w(\mathcal{D}/\mathcal{C})$ is fully faithful. Furthermore, if $K_0^{\text{cn}}(\mathcal{C}) \rightarrow K_0^{\text{cn}}(\mathcal{D})$ is injective, $\text{BGap}^w(\mathcal{D})/\text{BGap}^w(\mathcal{C}) \rightarrow \text{BGap}^w(\mathcal{D}/\mathcal{C})$ is an equivalence.*

Proof. From Lemma 4.17, we know $\text{Gap}^w(\mathcal{C})$ is a full subcategory of $\text{Gap}^w(\mathcal{D})$. For two objects $(X_{\bullet, \bullet}, Y_{\bullet, \bullet}), (X'_{\bullet, \bullet}, Y'_{\bullet, \bullet}) \in \text{BGap}^w(\mathcal{C})$, we have

$$\begin{aligned} \text{Map}_{\text{BGap}^w(\mathcal{C})}((X_{\bullet, \bullet}, Y_{\bullet, \bullet}), (X'_{\bullet, \bullet}, Y'_{\bullet, \bullet})) &\simeq \text{Map}_{\text{Gap}^w(\mathcal{C})}(X_{\bullet, \bullet}, X'_{\bullet, \bullet}) \times_{\text{Map}_{\text{colim}_n \mathcal{C}^n}(X_{\bullet, \bullet+1}, X'_{\bullet, \bullet+1})} \text{Map}_{\text{Gap}^w(\mathcal{C})}(Y_{\bullet, \bullet}, Y'_{\bullet, \bullet}) \\ &\simeq \text{Map}_{\text{Gap}^w(\mathcal{D})}(X_{\bullet, \bullet}, X'_{\bullet, \bullet}) \times_{\text{Map}_{\text{colim}_n \mathcal{D}^n}(X_{\bullet, \bullet+1}, X'_{\bullet, \bullet+1})} \text{Map}_{\text{Gap}^w(\mathcal{D})}(Y_{\bullet, \bullet}, Y'_{\bullet, \bullet}) \\ &\simeq \text{Map}_{\text{BGap}^w(\mathcal{D})}((X_{\bullet, \bullet}, Y_{\bullet, \bullet}), (X'_{\bullet, \bullet}, Y'_{\bullet, \bullet})) \end{aligned}$$

Hence, $\text{BGap}^w(\mathcal{C})$ is a full subcategory of $\text{BGap}^w(\mathcal{D})$. Now, the functor

$$\Phi : \text{BGap}^w(\mathcal{D}) \rightarrow \text{BGap}^w(\mathcal{D}/\mathcal{C})$$

induces a functor

$$\tilde{\Phi} : \text{BGap}^w(\mathcal{D})/\text{BGap}^w(\mathcal{C}) \rightarrow \text{BGap}^w(\mathcal{D}/\mathcal{C})$$

We need to prove it is an equivalence.

Denote by W the collection of morphisms in $\text{BGap}^w(\mathcal{D})$ whose cofibers are in $\text{BGap}^w(\mathcal{C})$. For two objects $(X_{\bullet, \bullet}, Y_{\bullet, \bullet}), (X'_{\bullet, \bullet}, Y'_{\bullet, \bullet}) \in \text{BGap}^w(\mathcal{D})$,

$$\begin{aligned} &\text{Map}_{\text{BGap}^w(\mathcal{D})/\text{BGap}^w(\mathcal{C})}((X_{\bullet, \bullet}, Y_{\bullet, \bullet}), (X'_{\bullet, \bullet}, Y'_{\bullet, \bullet})) \\ &\simeq \text{colim}_{(X''_{\bullet, \bullet}, Y''_{\bullet, \bullet}) \rightarrow (X_{\bullet, \bullet}, Y_{\bullet, \bullet}) \in W / (X_{\bullet, \bullet}, Y_{\bullet, \bullet})} \text{Map}_{\text{BGap}^w(\mathcal{D})}((X''_{\bullet, \bullet}, Y''_{\bullet, \bullet}), (X'_{\bullet, \bullet}, Y'_{\bullet, \bullet})). \end{aligned}$$

Notice that

$$\mathrm{Map}_{\mathrm{BGap}^w(\mathcal{D})}((X''_{\bullet,\bullet}, Y''_{\bullet,\bullet}), (X'_{\bullet,\bullet}, Y'_{\bullet,\bullet})) \simeq \mathrm{Map}_{\mathrm{Gap}^w(\mathcal{D})}(X''_{\bullet,\bullet}, X'_{\bullet,\bullet}) \times_{\mathrm{Map}_{\mathrm{colim}_n \mathcal{D}^n}(X''_{\bullet,\bullet+1}, X'_{\bullet,\bullet+1})} \mathrm{Map}_{\mathrm{Gap}^w(\mathcal{D})}(Y''_{\bullet,\bullet}, Y'_{\bullet,\bullet})$$

From Lemma 4.18, we know $\tilde{\phi} : \mathrm{Gap}^w(\mathcal{D})/\mathrm{Gap}^w(\mathcal{C}) \rightarrow \mathrm{Gap}^w(\mathcal{D}/\mathcal{C})$ is an equivalence. Hence,

$$\begin{aligned} \mathrm{colim}_{(X''_{\bullet,\bullet}, Y''_{\bullet,\bullet}) \rightarrow (X_{\bullet,\bullet}, Y_{\bullet,\bullet}) \in W/(X_{\bullet,\bullet}, Y_{\bullet,\bullet})} \mathrm{Map}_{\mathrm{Gap}^w(\mathcal{D})}(X''_{\bullet,\bullet}, X'_{\bullet,\bullet}) &\simeq \mathrm{Map}_{\mathrm{Gap}^w(\mathcal{D}/\mathcal{C})}(\tilde{\phi}(X_{\bullet,\bullet}), \tilde{\phi}(X'_{\bullet,\bullet})), \\ \mathrm{colim}_{(X''_{\bullet,\bullet}, Y''_{\bullet,\bullet}) \rightarrow (X_{\bullet,\bullet}, Y_{\bullet,\bullet}) \in W/(X_{\bullet,\bullet}, Y_{\bullet,\bullet})} \mathrm{Map}_{\mathrm{Gap}^w(\mathcal{D})}(Y''_{\bullet,\bullet}, Y'_{\bullet,\bullet}) &\simeq \mathrm{Map}_{\mathrm{Gap}^w(\mathcal{D}/\mathcal{C})}(\tilde{\phi}(Y_{\bullet,\bullet}), \tilde{\phi}(Y'_{\bullet,\bullet})), \\ \mathrm{colim}_{(X''_{\bullet,\bullet}, Y''_{\bullet,\bullet}) \rightarrow (X_{\bullet,\bullet}, Y_{\bullet,\bullet}) \in W/(X_{\bullet,\bullet}, Y_{\bullet,\bullet})} \mathrm{Map}_{\mathrm{colim}_n \mathcal{D}^n}(X''_{\bullet,\bullet+1}, X'_{\bullet,\bullet+1}) &\simeq \mathrm{Map}_{\mathrm{colim}_n(\mathcal{D}/\mathcal{C})^n}(X_{\bullet,\bullet+1}, X'_{\bullet,\bullet+1}) \end{aligned}$$

It means

$$\begin{aligned} &\mathrm{Map}_{\mathrm{BGap}^w(\mathcal{D})/\mathrm{BGap}^w(\mathcal{C})}((X_{\bullet,\bullet}, Y_{\bullet,\bullet}), (X'_{\bullet,\bullet}, Y'_{\bullet,\bullet})) \\ &\simeq \mathrm{Map}_{\mathrm{Gap}^w(\mathcal{D}/\mathcal{C})}(\tilde{\phi}(X_{\bullet,\bullet}), \tilde{\phi}(X'_{\bullet,\bullet})) \times_{\mathrm{Map}_{\mathrm{colim}_n(\mathcal{D}/\mathcal{C})^n}(X_{\bullet,\bullet+1}, X'_{\bullet,\bullet+1})} \mathrm{Map}_{\mathrm{Gap}^w(\mathcal{D}/\mathcal{C})}(\tilde{\phi}(Y_{\bullet,\bullet}), \tilde{\phi}(Y'_{\bullet,\bullet})), \\ &\simeq \mathrm{Map}_{\mathrm{BGap}^w(\mathcal{D}/\mathcal{C})}(\tilde{\Phi}(X_{\bullet,\bullet}, Y_{\bullet,\bullet}), \tilde{\Phi}(X'_{\bullet,\bullet}, Y'_{\bullet,\bullet})) \end{aligned}$$

i.e. $\tilde{\Phi}$ is fully faithful.

Now we assume $K_0^{\mathrm{cn}}(\mathcal{C}) \rightarrow K_0^{\mathrm{cn}}(\mathcal{D})$ is injective. Recall that $\phi : \mathrm{Gap}^w(\mathcal{D}) \rightarrow \mathrm{Gap}^w(\mathcal{D}/\mathcal{C})$ is essentially surjective. For $(F_{\bullet,\bullet}, G_{\bullet,\bullet}) \in \mathrm{BGap}^w(\mathcal{D}/\mathcal{C})$, there are $X_{\bullet,\bullet}, Y_{\bullet,\bullet} \in \mathrm{Gap}^w(\mathcal{D})$ such that $\phi(X_{\bullet,\bullet}) \simeq F_{\bullet,\bullet}, \phi(Y_{\bullet,\bullet}) \simeq G_{\bullet,\bullet}$.

Assume $X_{0,i} = Y_{0,i} = 0$ for $i > n$. Define two sequences

$$\begin{aligned} 0 &\rightarrow X'_{0,1} \rightarrow \cdots \rightarrow X'_{0,n} \rightarrow X'_{0,n+1} \\ 0 &\rightarrow Y'_{0,1} \rightarrow \cdots \rightarrow Y'_{0,n} \rightarrow Y'_{0,n+1} \end{aligned}$$

as follows:

- Since $X_{0,1} \simeq Y_{0,1}$ in $\mathcal{D}/\mathcal{C}$, there are two morphisms $Z_1 \rightarrow X_{0,1}, Z_1 \rightarrow Y_{0,1}$ whose cofibers are in $\mathcal{C}$. Define $X'_{0,1} = Y'_{0,1} := Z_{0,1}$.
- Consider the map $Z_1 \rightarrow X_{0,1} \rightarrow X_{0,2}$. Since $\mathrm{cofib}(Z_1 \rightarrow X_{0,1}) \in \mathcal{C}$, $\mathrm{cofib}(Z_1 \rightarrow X_{0,2}) \simeq X_{1,2}$ in $\mathcal{D}/\mathcal{C}$. Hence,

$$\mathrm{cofib}(Z_1 \rightarrow X_{0,2}) \simeq X_{1,2} \simeq Y_{1,2} \simeq \mathrm{cofib}(Z_1 \rightarrow Y_{0,2}) \in \mathcal{D}/\mathcal{C}.$$

There are two morphisms $Z_2 \rightarrow \mathrm{cofib}(Z_1 \rightarrow X_{0,2}), Z_2 \rightarrow \mathrm{cofib}(Z_1 \rightarrow Y_{0,2})$ whose cofibers are in $\mathcal{C}$. Define $X'_{0,2} := Z_2 \times_{\mathrm{cofib}(Z_1 \rightarrow X_{0,2})} X_{0,2}$ and $Y'_{0,2} := Z_2 \times_{\mathrm{cofib}(Z_1 \rightarrow Y_{0,2})} Y_{0,2}$. Then by Lemma 1.6,

$$\begin{aligned} \mathrm{fib}(X'_{0,2} \rightarrow Z_2) &\simeq \mathrm{fib}(X_{0,2} \rightarrow \mathrm{cofib}(Z_1 \rightarrow X_{0,2})) \simeq Z_1 \\ \mathrm{fib}(Y'_{0,2} \rightarrow Z_2) &\simeq \mathrm{fib}(Y_{0,2} \rightarrow \mathrm{cofib}(Z_1 \rightarrow Y_{0,2})) \simeq Z_1 \end{aligned}$$

Then we get $X'_{0,1} = Z_1 \rightarrow X'_{0,2}$ and $Y'_{0,1} = Z_1 \rightarrow Y'_{0,2}$ whose cofibers are both Z_2 .

- For $2 \leq i \leq n$, consider the map $X'_{0,i} \rightarrow X_{0,i} \rightarrow X_{0,i+1}$. By Lemma 1.6, $\mathrm{cofib}(X'_{0,i} \rightarrow X_{0,i}) \simeq \mathrm{cofib}(Z_i \rightarrow \mathrm{cofib}(X'_{0,i-1} \rightarrow X_{0,i}))$ is in $\mathcal{C}$. We have

$$\mathrm{cofib}(X'_{0,i} \rightarrow X_{0,i+1}) \simeq X_{i,i+1} \simeq Y_{i,i+1} \simeq \mathrm{cofib}(Y'_{0,i} \rightarrow Y_{0,i+1}) \in \mathcal{D}/\mathcal{C}.$$

There are two morphisms $Z_{i+1} \rightarrow \mathrm{cofib}(X'_{0,i} \rightarrow X_{0,i+1}), Z_{i+1} \rightarrow \mathrm{cofib}(Y'_{0,i} \rightarrow Y_{0,i+1})$ whose cofibers are in $\mathcal{C}$. Define $X'_{0,i+1} := Z_{i+1} \times_{\mathrm{cofib}(X'_{0,i} \rightarrow X_{0,i+1})} X_{0,i+1}$ and $Y'_{0,i+1} := Z_{i+1} \times_{\mathrm{cofib}(Y'_{0,i} \rightarrow Y_{0,i+1})} Y_{0,i+1}$. Then by Lemma 1.6,

$$\begin{aligned} \mathrm{fib}(X'_{0,i+1} \rightarrow Z_{i+1}) &\simeq \mathrm{fib}(X_{0,i+1} \rightarrow \mathrm{cofib}(X'_{0,i} \rightarrow X_{0,i+1})) \simeq X'_{0,i} \\ \mathrm{fib}(Y'_{0,i+1} \rightarrow Z_{i+1}) &\simeq \mathrm{fib}(Y_{0,i+1} \rightarrow \mathrm{cofib}(Y'_{0,i} \rightarrow Y_{0,i+1})) \simeq Y'_{0,i} \end{aligned}$$

Then we get $X'_{0,i} \rightarrow X'_{0,i+1}$ and $Y'_{0,i} \rightarrow Y'_{0,i+1}$ whose cofibers are both Z_{i+1} .

The same as above, $\text{cofib}(X'_{0,n+1} \rightarrow X_{0,n+1} = 0)$ is in $\mathcal{C}$. Hence, $X'_{0,n+1} \in \mathcal{C}$. Also, $Y'_{0,n+1} \in \mathcal{C}$.

Since $[X'_{0,n+1}] = [Z_1] + [Z_2] + \cdots + [Z_{n+1}] = [Y'_{0,n+1}]$ in $K_0^{\text{cn}}(\mathcal{D})$, we also have $[X'_{0,n+1}] = [Y'_{0,n+1}]$ in $K_0^{\text{cn}}(\mathcal{C})$ since $K_0^{\text{cn}}(\mathcal{C}) \rightarrow K_0^{\text{cn}}(\mathcal{D})$ is injective. Hence, there are two factorizations of $X'_{0,n+1} \rightarrow 0$ and $Y'_{0,n+1} \rightarrow 0$ in $\mathcal{C}$

$$\begin{aligned} X'_{0,n+1} &\rightarrow X'_{0,n+2} \rightarrow \cdots \rightarrow X'_{0,n+m} = 0 \\ Y'_{0,n+1} &\rightarrow Y'_{0,n+2} \rightarrow \cdots \rightarrow Y'_{0,n+m} = 0 \end{aligned}$$

such that $\text{cofib}(X'_{0,n+i} \rightarrow X'_{0,n+i+1}) \simeq \text{cofib}(Y'_{0,n+i} \rightarrow Y'_{0,n+i+1})$ for some $m > 0$.

Now we get two objects in $\text{Gap}^w(\mathcal{D})$:

$$\begin{aligned} 0 &\rightarrow X'_{0,1} \rightarrow \cdots \rightarrow X'_{0,n} \rightarrow X'_{0,n+1} \rightarrow \cdots \rightarrow X'_{0,n+m} \rightarrow 0 \rightarrow \cdots \\ 0 &\rightarrow Y'_{0,1} \rightarrow \cdots \rightarrow Y'_{0,n} \rightarrow Y'_{0,n+1} \rightarrow \cdots \rightarrow Y'_{0,n+m} \rightarrow 0 \rightarrow \cdots \end{aligned}$$

Above discussion tells us $(X'_{\bullet,\bullet}, Y'_{\bullet,\bullet}) \in \text{BGap}^w(\mathcal{D})$.

There are natural morphisms $X'_{0,i} \rightarrow X_{0,i}$ (when $i > n$, they are just $X'_{0,i} \rightarrow 0$), and their cofibers are in $\mathcal{C}$. The same for $Y'_{0,i} \rightarrow Y_{0,i}$. They give morphisms

$$X'_{\bullet,\bullet} \rightarrow X_{\bullet,\bullet}, Y'_{\bullet,\bullet} \rightarrow Y_{\bullet,\bullet}$$

whose cofibers are in $\text{Gap}^w(\mathcal{C})$. Hence, $\phi(X'_{\bullet,\bullet}) \simeq \phi(X_{\bullet,\bullet}) \simeq F_{\bullet,\bullet}$ and $\phi(Y'_{\bullet,\bullet}) \simeq \phi(Y_{\bullet,\bullet}) \simeq G_{\bullet,\bullet}$. Thus,

$$\Phi(X'_{\bullet,\bullet}, Y'_{\bullet,\bullet}) \simeq (\phi(X'_{\bullet,\bullet}), \phi(Y'_{\bullet,\bullet})) \simeq (F_{\bullet,\bullet}, G_{\bullet,\bullet}).$$

It tells us Φ is essential surjective, i.e. so is $\tilde{\Phi}$.

In conclusion, $\tilde{\Phi}$ is an equivalence. □

Lemma 4.22. *Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. Then*

$$\begin{aligned} \text{Gap}([n], \text{BGap}^w(\mathcal{F})) &\simeq \text{BGap}^w(\text{Gap}([n], \mathcal{F})) \\ \text{Gap}^w([n], \text{BGap}^w(\mathcal{F})) &\simeq \text{BGap}^w(\text{Gap}^w([n], \mathcal{F})) \end{aligned}$$

where $\text{BGap}^w(\mathcal{F})$ is the functor $\text{BGap}^w(\mathcal{A}) \rightarrow \text{BGap}^w(\mathcal{C})$ induced by $\mathcal{C}$.

Proof. Recall BGap^w is left exact from Lemma 4.19. Then

$$\begin{aligned} \text{BGap}^w(\text{Gap}([n], \mathcal{F})) &\simeq \text{BGap}^w(\text{Gap}([n], \mathcal{C}) \times_{\mathcal{C}^n} \mathcal{A}^n) \\ &\simeq \text{BGap}^w(\text{Gap}([n], \mathcal{C})) \times_{\text{BGap}^w(\mathcal{C})^n} \text{BGap}^w(\mathcal{A})^n \end{aligned}$$

From Lemma 4.20, $\text{BGap}^w(\text{Gap}([n], \mathcal{C})) \simeq \text{Gap}([n], \text{BGap}^w(\mathcal{C}))$. Hence,

$$\begin{aligned} \text{BGap}^w(\text{Gap}([n], \mathcal{F})) &\simeq \text{Gap}([n], \text{BGap}^w(\mathcal{C})) \times_{\text{BGap}^w(\mathcal{C})^n} \text{BGap}^w(\mathcal{A})^n \\ &\simeq \text{Gap}([n], \text{BGap}^w(\mathcal{F})). \end{aligned}$$

Similar, we have

$$\text{Gap}^w([n], \text{BGap}^w(\mathcal{F})) \simeq \text{BGap}^w(\text{Gap}^w([n], \mathcal{F})).$$

□

Lemma 4.23. *Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_{\infty}^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. Then $K_0(\text{Gap}^w(\mathcal{F})) \rightarrow K_0(\text{Gap}(\mathcal{F}))$ is injective if $K_0(Q(\mathcal{F})) = 0$.*

Proof. By Proposition 3.2, Theorem 3.9 and the split cofiber sequence

$$\mathcal{U}_{\text{loc}}(\text{colim}_n \mathcal{A}^{n-1}) \xrightarrow{\mathcal{U}_{\text{loc}}(\tau_{\mathcal{A}})} \mathcal{U}_{\text{loc}}(\text{Gap}^w(\mathcal{F})) \rightarrow \mathcal{U}_{\text{loc}}(Q(\mathcal{F})),$$

we know $K_0(\text{Gap}^w(\mathcal{F})) \rightarrow K_0(\text{Gap}(\mathcal{F}))$ is just

$$\text{colim}_n K_0(\mathcal{A})^{n-1} \oplus K_0(Q(\mathcal{F})) \rightarrow \text{colim}_n K_0(\mathcal{A})^{n-1} \oplus K_0(\mathcal{A}).$$

Hence, if $K_0(Q(\mathcal{F})) = 0$, $K_0(\text{Gap}^w(\mathcal{F})) \rightarrow K_0(\text{Gap}(\mathcal{F}))$ is injective. □

Proposition 4.24. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ satisfies the dévissage condition and $K_0(Q(\mathcal{F})) = 0$, $\text{BGap}^w(\mathcal{F})$ satisfies the dévissage condition.

Proof. From Proposition 2.9, we have $E(\mathcal{F}) = \mathcal{C}$ and we only need to prove

$$E(\text{BGap}^w(\mathcal{F})) \simeq \text{BGap}^w(E(\mathcal{F})).$$

By Lemma 4.21 and Lemma 4.22, we have

$$\begin{aligned} E(\text{BGap}^w(\mathcal{F})) &\simeq \text{Gap}(\text{BGap}^w(\mathcal{F}))/\text{Gap}^w(\text{BGap}^w(\mathcal{F})) \\ &\simeq \text{BGap}^w(\text{Gap}(\mathcal{F}))/\text{BGap}^w(\text{Gap}^w(\mathcal{F})) \end{aligned}$$

Since $\text{BGap}^w(\text{Gap}(\mathcal{F}))/\text{Gap}^w(\mathcal{F}) \simeq \text{BGap}^w(E(\mathcal{F}))$, to prove $E(\text{BGap}^w(\mathcal{F})) \simeq \text{BGap}^w(E(\mathcal{F}))$, we need to prove

$$\text{BGap}^w(\text{Gap}(\mathcal{F}))/\text{BGap}^w(\text{Gap}^w(\mathcal{F})) \rightarrow \text{BGap}^w(\text{Gap}(\mathcal{F})/\text{Gap}^w(\mathcal{F}))$$

is an equivalence. From Lemma 4.21 and Lemma 4.23, we just need $K_0(Q(\mathcal{F})) = 0$. $\square$

Proposition 4.24 and Theorem 4.4 tells us:

Corollary 4.25. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be a 1-fillable exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ satisfies the dévissage condition, $\text{BGap}(\mathcal{F})$ satisfies the dévissage condition.

Lemma 4.26.

$$\text{BGap}^w(\text{Gap}^\vee(\mathcal{F})) \simeq \text{Gap}^\vee(\text{BGap}^w(\mathcal{F})).$$

Proof. By Lemma 4.19 and Lemma 4.22,

$$\begin{aligned} \text{BGap}^w(\text{Gap}^\vee(\mathcal{F})) &\simeq \text{BGap}^w(\text{Gap}(\mathcal{F})) \times_{\text{BGap}^w(\mathcal{C})} \text{BGap}^w(\text{Gap}(\mathcal{F})) \\ &\simeq \text{Gap}(\text{BGap}^w(\mathcal{F})) \times_{\text{BGap}^w(\mathcal{C})} \text{Gap}(\text{BGap}^w(\mathcal{F})) \\ &\simeq \text{Gap}^\vee(\text{BGap}^w(\mathcal{F})). \end{aligned}$$

$\square$

Definition 4.27. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. $\mathcal{F}$ is called *strongly 1-fillable* if there is an exact functor $\text{fill} : \text{Gap}^\vee(\mathcal{F}) \rightarrow \text{Gap}(\text{Gap}_{\mathcal{F}}^\vee)$ such that $\text{ev}_{\text{Gap}_{\mathcal{F}}^\vee} \circ \text{fill} \simeq \text{id}$.

Remark 4.28. Notice that $\mathcal{F}$ is 1-fillable if and only if $\text{ev}_{\text{Gap}_{\mathcal{F}}^\vee} : \text{Gap}(\text{Gap}_{\mathcal{F}}^\vee) \rightarrow \text{Gap}^\vee(\mathcal{F})$ is essentially surjective (Corollary 2.14). Hence, this condition is naturally stronger than 1-fillability. Hence, the above results of 1-fillability are also true.

Theorem 4.29. Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be an exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ is strongly 1-fillable, $\text{BGap}^w(\mathcal{F})$ is also strongly 1-fillable.

Proof. Lemma 4.22 and Lemma 4.26 tells us

$$\text{Gap}(\text{Gap}_{\text{BGap}^w(\mathcal{F})}^\vee) \simeq \text{BGap}^w(\text{Gap}(\text{Gap}_{\mathcal{F}}^\vee)).$$

Hence,

$$\text{BGap}^w(\text{fill})\text{Gap}^\vee(\text{BGap}^w(\mathcal{F})) \rightarrow \text{Gap}(\text{Gap}_{\text{BGap}^w(\mathcal{F})}^\vee)$$

satisfies $\text{BGap}^w(\text{ev}_{\text{Gap}_{\mathcal{F}}^\vee}) \circ \text{BGap}^w(\text{fill}) \simeq \text{id}$. Notice that $\text{BGap}^w(\text{ev}_{\text{Gap}_{\mathcal{F}}^\vee}) \simeq \text{ev}_{\text{Gap}_{\text{BGap}^w(\mathcal{F})}^\vee}$ naturally, we have $\text{ev}_{\text{Gap}_{\text{BGap}^w(\mathcal{F})}^\vee} \circ \text{BGap}^w(\text{fill}) \simeq \text{id}$. Thus, $\text{BGap}^w(\mathcal{F})$ is a strongly 1-fillable. $\square$

Theorem 4.30. (Theorem C) Let $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{C} \in \text{Cat}_\infty^{\text{perf}}$ be a strongly 1-fillable exact functor between small stable idempotent-complete ∞ -categories. If $\mathcal{F}$ satisfies the dévissage condition, it induces isomorphisms $K_n(\mathcal{A}) \xrightarrow{\cong} K_n(\mathcal{C})$ for all $n \geq 0$.

Proof. We prove it by induction on n .

- $n = 0$. It is Theorem 4.4.
- Assume it holds for $n - 1$. From Lemma 4.24, Corollary 4.25 and assumption, $K_{n-1}(\text{BGap}^w(\mathcal{A})) \rightarrow K_{n-1}(\text{BGap}^w(\mathcal{C}))$ is an isomorphism. In the last section, we know that the cofiber sequence

$$K(\text{colim}_n \mathcal{A}^{n-1}) \rightarrow K(\text{BGap}^w(\mathcal{A})) \rightarrow K(\theta(\mathcal{A}))$$

split. Hence, $K_{n-1}(\text{BGap}^w(\mathcal{A})) \simeq K_{n-1}(\text{colim}_n \mathcal{A}^{n-1}) \oplus K_{n-1}(\theta(\mathcal{A}))$. Since $K_{n-1}(\text{BGap}^w(\mathcal{A})) \rightarrow K_{n-1}(\text{BGap}^w(\mathcal{C}))$ is injective, both $K_{n-1}(\text{colim}_n \mathcal{A}^{n-1}) \rightarrow K_{n-1}(\text{colim}_n \mathcal{C}^{n-1})$ and $K_{n-1}(\theta(\mathcal{A})) \rightarrow K_{n-1}(\theta(\mathcal{C}))$ is an isomorphism. By Theorem 3.14, it means $K_n(\mathcal{A}) \rightarrow K_n(\mathcal{C})$ is isomorphism.

□

5 ON THE THEOREM OF HEART

Recall the theorem of heart.

Theorem 5.1. [Bar15] *Let $\mathcal{C}$ be a stable ∞ -category equipped with a bounded t -structure. Then the natural inclusion $\mathcal{C}^\heartsuit \rightarrow \mathcal{C}$ induces isomorphisms $K_n(\mathcal{C}^\heartsuit) \rightarrow K_n(\mathcal{C})$ for all $n \geq 0$.*

t -structure is firstly introduced by [BBD82, Definition 1.3.1]. [Lur17, Definition 1.2.1.1] gives the definition of t -structure on stable ∞ -categories with homological indexing.

Lemma 5.2. *Let J be a finite ∞ -category and $\mathcal{E}$ be a small abelian category. Then there is a natural equivalence*

$$\mathcal{D}^b(\text{Fun}(J, \mathcal{E})) \simeq \text{Fun}(J, \mathcal{D}^b(\mathcal{E})).$$

Proof. There is a natural functor

$$\mathcal{D}^b(\text{Fun}(J, \mathcal{E})) \rightarrow \text{Fun}(J, \mathcal{D}^b(\mathcal{E})).$$

We only need to prove it induces equivalence

$$\text{Fun}^{\text{ex}}(\mathcal{D}^b(\text{Fun}(J, \mathcal{E})), \mathcal{C}) \simeq \text{Fun}^{\text{ex}}(\text{Fun}(J, \mathcal{D}^b(\mathcal{E})), \mathcal{C})$$

for any stable ∞ -categories $\mathcal{C}$. By [Kle22, Theorem 1], we know

$$\begin{aligned} \text{Fun}^{\text{ex}}(\mathcal{D}^b(\text{Fun}(J, \mathcal{E})), \mathcal{C}) &\simeq \text{Fun}^{\text{ex}}(\text{Fun}(J, \mathcal{E}), \mathcal{C}) \\ &\simeq \text{Fun}(J, \text{Fun}^{\text{ex}}(\mathcal{E}, \mathcal{C})) \\ &\simeq \text{Fun}(J, \text{Fun}^{\text{ex}}(\mathcal{D}^b(\mathcal{E}), \mathcal{C})) \\ &\simeq \text{Fun}^{\text{ex}}(\text{Fun}(J, \mathcal{D}^b(\mathcal{E})), \mathcal{C}). \end{aligned}$$

□

Remark 5.3. For the details of derived category of abelian category, there are some good references [Lur16, Appendix C][Kra14].

Definition 5.4. Let $\mathcal{E}$ be a small abelian category. Define $\text{Seq}(\mathcal{E}) := \text{colim}_n \text{Fun}([n], \mathcal{E})$ and $\text{Seq}^\vee(\mathcal{E}) := \text{colim}_n \text{Fun}([n] \sqcup_{[0]} [n], \mathcal{E})$, where the pushout is defined by $[0] \rightarrow [n], 0 \mapsto n$ and the filtered colimits are defined by $[n+1] \rightarrow [n], n+1 \mapsto n$.

Corollary 5.5. There are equivalences:

$$\mathcal{D}^b(\text{Seq}(\mathcal{E})) \simeq \text{Gap}(\mathcal{D}^b(\mathcal{E})), \mathcal{D}^b(\text{Seq}^\vee(\mathcal{E})) \simeq \text{Gap}^\vee(\mathcal{D}^b(\mathcal{E}))$$

Denote by $\Phi : \mathcal{D}^b(\mathcal{C}^\heartsuit) \rightarrow \mathcal{C}$ the canonical t -exact functor constructed by [AGH19, Proposition 3.26].

Lemma 5.6. $\Phi : \mathcal{D}^b(\mathcal{C}^\heartsuit) \rightarrow \mathcal{C}$ satisfies the dévissage condition.

Proof. It is obvious that $\mathcal{C}^\heartsuit \subseteq E(\Phi)$, which forces $E(\Phi) = \mathcal{C}$ by boundness of the t-structure. By Proposition 2.9, Φ satisfies the dévissage condition. $\square$

Lemma 5.7. *Let $\mathcal{C}$ be a stable ∞ -category equipped with a bounded t-structure. Then $\text{Gap}^\vee(\mathcal{C})$ admits a bounded t-structure with heart $\text{Seq}^\vee(\mathcal{C}^\heartsuit)$.*

Proof. Define full subcategories:

$$\begin{aligned}\text{Gap}^\vee(\mathcal{C})_{\geq 0} &:= \{(X, Y) \in \text{Gap}^\vee(\mathcal{C}) \mid X_{0,n}, Y_{0,n} \in \mathcal{C}_{\geq 0}, \forall n \geq 0\} \\ \text{Gap}^\vee(\mathcal{C})_{\leq 0} &:= \{(X, Y) \in \text{Gap}^\vee(\mathcal{C}) \mid X_{0,n}, Y_{0,n} \in \mathcal{C}_{\leq 0}, \forall n \geq 0\}.\end{aligned}$$

- $\text{Gap}^\vee(\mathcal{C})_{\geq 1} \subseteq \text{Gap}^\vee(\mathcal{C})_{\geq 0}$ and $\text{Gap}^\vee(\mathcal{C})_{\leq 0} \subseteq \text{Gap}^\vee(\mathcal{C})_{\leq 1}$ are obvious.
- let $(X_{\bullet,\bullet}, Y_{\bullet,\bullet}) \in \text{Gap}^\vee(\mathcal{C})_{\geq 0}$ and $(X'_{\bullet,\bullet}, Y'_{\bullet,\bullet}) \in \text{Gap}^\vee(\mathcal{C})_{\leq -1}$. To prove $\text{Map}((X_{\bullet,\bullet}, Y_{\bullet,\bullet}), (X'_{\bullet,\bullet}, Y'_{\bullet,\bullet})) \simeq 0$, we just need to prove $\text{Map}(X_{\bullet,\bullet}, X'_{\bullet,\bullet}) \simeq 0$. Since $\text{Map}_{\mathcal{C}}(X_{0,n}, X'_{0,n}) \simeq 0$ and $\text{Map}_{\mathcal{C}}(X_{0,n}[1], X'_{0,n+1}) \simeq 0$ for all $n \geq 0$, we know $\text{Map}(X_{\bullet,\bullet}, X'_{\bullet,\bullet}) = 0$.
- let $(X_{\bullet,\bullet}, Y_{\bullet,\bullet}) \in \text{Gap}^\vee(\mathcal{C})$, define

$$\begin{aligned}\tau_{\geq n}(X_{\bullet,\bullet}, Y_{\bullet,\bullet}) &= (0 \rightarrow \tau_{\geq n}X_{0,1} \rightarrow \tau_{\geq n}X_{0,2} \rightarrow \cdots, 0 \rightarrow \tau_{\geq n}Y_{0,1} \rightarrow \tau_{\geq n}Y_{0,2} \rightarrow \cdots) \\ \tau_{\leq n}(X_{\bullet,\bullet}, Y_{\bullet,\bullet}) &= (0 \rightarrow \tau_{\leq n}X_{0,1} \rightarrow \tau_{\leq n}X_{0,2} \rightarrow \cdots, 0 \rightarrow \tau_{\leq n}Y_{0,1} \rightarrow \tau_{\leq n}Y_{0,2} \rightarrow \cdots).\end{aligned}$$

$\square$

Lemma 5.8.

- (1) For $X \in \mathcal{D}^b(\mathcal{C}^\heartsuit)$, if $\Phi(X) \in \mathcal{C}_{[a,b]}$, then $X \in \mathcal{D}^b(\mathcal{C}^\heartsuit)_{[a,b]}$.
- (2) For $X \in \mathcal{D}^b(\mathcal{C}^\heartsuit)_{\geq 0}, Y \in \mathcal{D}^b(\mathcal{C}^\heartsuit)_{\leq 0}$, $\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, Y) \simeq \text{Map}_{\mathcal{C}^\heartsuit}(\pi_0 X, \pi_0 Y) \simeq \text{Map}_{\mathcal{C}}(\Phi(X), \Phi(Y))$.
- (3) Let $X \rightarrow Y \rightarrow Z$ be a cofiber sequence in $\mathcal{C}$. Then there is a cofiber sequence

$$\text{cofib}(\tau_{\geq n}X \rightarrow \tau_{\geq n}Y) \rightarrow \tau_{\geq n}Z \rightarrow \pi_n Z / \text{coker}(\pi_n X \rightarrow \pi_n Y)[n].$$

Proof.

- (1) It is because for every n , $\pi_n X \rightarrow \pi_n \Phi(X)$ is equivalence.
- (2) Apply the (co)fiber sequence $\pi_0 Y \rightarrow Y \rightarrow \tau_{\leq -1}Y$, we have fiber sequence

$$\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, \pi_0 Y) \rightarrow \text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, Y) \rightarrow \text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, \tau_{\leq -1}Y).$$

Since $\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, \tau_{\leq -1}Y)$ is contractible, we have equivalence

$$\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, \pi_0 Y) \simeq \text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, Y).$$

Apply the (co)fiber sequence $\tau_{\geq 1}X \rightarrow X \rightarrow \pi_0 X$, we have fiber sequence

$$\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(\tau_{\geq 1}X, \pi_0 Y) \rightarrow \text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, \pi_0 Y) \rightarrow \text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(\pi_0 X, \pi_0 Y)$$

Since $\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(\tau_{\geq 1}X, \pi_0 Y)$ is contractible, we have equivalence

$$\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, \pi_0 Y) \simeq \text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(\pi_0 X, \pi_0 Y).$$

Hence, $\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(X, Y) \simeq \text{Map}_{\mathcal{C}^\heartsuit}(\pi_0 X, \pi_0 Y)$. Similarly,

$$\text{Map}_{\mathcal{C}}(\Phi(X), \Phi(Y)) \simeq \text{Map}_{\mathcal{C}^\heartsuit}(\pi_0 \Phi(X), \pi_0 \Phi(Y))$$

- (3) Compare two long exact sequences

$$\begin{array}{ccccccccc}\pi_i \tau_{\geq n}X & \longrightarrow & \pi_i \tau_{\geq n}Y & \longrightarrow & \pi_i \text{cofib}(\tau_{\geq n}X \rightarrow \tau_{\geq n}Y) & \longrightarrow & \pi_{i-1} \tau_{\geq n}X & \longrightarrow & \pi_{i-1} \tau_{\geq n}Y \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ \pi_i X & \longrightarrow & \pi_i Y & \longrightarrow & \pi_i Z & \longrightarrow & \pi_{i-1} X & \longrightarrow & \pi_{i-1} Y\end{array}$$

If $i > n$, by the five lemma, we know $\pi_i \text{cofib}(\tau_{\geq n}X \rightarrow \tau_{\geq n}Y) \rightarrow \pi_i Z$ is equivalence.

Since $\pi_i \tau_{\geq n}X = \pi_i \tau_{\geq n}Y = 0$ when $i < n$, we know $\pi_i \text{cofib}(\tau_{\geq n}X \rightarrow \tau_{\geq n}Y) = 0$ for all $i < n$. Besides, $\pi_n \text{cofib}(\tau_{\geq n}X \rightarrow \tau_{\geq n}Y) = \text{coker}(\pi_n X \rightarrow \pi_n Y)$ and $\pi_n \text{cofib}(\tau_{\geq n}X \rightarrow \tau_{\geq n}Y) \rightarrow \pi_n Z$ is monomorphism. Hence,

$$\text{cofib}(\text{cofib}(\tau_{\geq n}X \rightarrow \tau_{\geq n}Y) \rightarrow \tau_{\geq n}Z) \simeq \pi_n Z / \text{coker}(\pi_n X \rightarrow \pi_n Y)[n].$$

□

Proposition 5.9. Let $\mathcal{C}$ be a stable ∞ -category equipped with a bounded t-structure. Then $\text{Gap}^\vee(\Phi)$ admits a bounded t-structure with heart $\text{Seq}^\vee(\mathcal{C}^\heartsuit)$.

Proof. Define full subcategories

$$\begin{aligned}\text{Gap}^\vee(\Phi)_{\geq 0} &= \{(A_\bullet, X_{\bullet,\bullet}, B_\bullet, Y_{\bullet,\bullet}) \in \text{Gap}^\vee(\Phi) \mid X_{0,n}, Y_{0,n} \in \mathcal{C}_{\geq 0}, \forall n \geq 0\} \\ \text{Gap}^\vee(\Phi)_{\leq 0} &= \{(A_\bullet, X_{\bullet,\bullet}, B_\bullet, Y_{\bullet,\bullet}) \in \text{Gap}^\vee(\Phi) \mid X_{0,n}, Y_{0,n} \in \mathcal{C}_{\leq 0}, \forall n \geq 0\}\end{aligned}$$

- $\text{Gap}^\vee(\Phi)_{\geq 1} \subseteq \text{Gap}^\vee(\Phi)_{\geq 0}$ and $\text{Gap}^\vee(\Phi)_{\leq 0} \subseteq \text{Gap}^\vee(\Phi)_{\leq 1}$ are obvious.
- Let $(A_\bullet, X_{\bullet,\bullet}, B_\bullet, Y_{\bullet,\bullet}) \in \text{Gap}^\vee(\Phi)_{\geq 0}$ and $(A'_\bullet, X'_{\bullet,\bullet}, B'_\bullet, Y'_{\bullet,\bullet}) \in \text{Gap}^\vee(\Phi)_{\leq -1}$. To prove

$$\text{Map}((A_\bullet, X_{\bullet,\bullet}, B_\bullet, Y_{\bullet,\bullet}), (A'_\bullet, X'_{\bullet,\bullet}, B'_\bullet, Y'_{\bullet,\bullet})) = 0,$$

we just need to prove $\text{Map}((A_\bullet, X_{\bullet,\bullet}), (A'_\bullet, X'_{\bullet,\bullet})) = 0$. It is

$$\text{Map}((A_\bullet, X_{\bullet,\bullet}), (A'_\bullet, X'_{\bullet,\bullet})) \simeq \text{Map}(X_{\bullet,\bullet}, X'_{\bullet,\bullet}) \times_{\text{colim}_n \prod_{i=1}^n \text{Map}_{\mathcal{C}}(A_i, A'_i)} \text{colim}_n \prod_{i=1}^n \text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(A_i, A'_i)$$

Since $\text{Map}(X_{\bullet,\bullet}, X'_{\bullet,\bullet}) \simeq 0$, we only need to prove

$$\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(A_n, A'_n) \rightarrow \text{Map}_{\mathcal{C}}(A_n, A'_n)$$

is equivalence for each $n \geq 1$.

Since $\Phi(A_n) = \text{cofib}(X_{0,n-1} \rightarrow X_{0,n})$ and $\Phi(A'_n) = \text{cofib}(X'_{0,n-1} \rightarrow X'_{0,n})$, we know $\Phi(A_n) \in \mathcal{C}_{\geq 0}$, $\Phi(A'_n) \in \mathcal{C}_{\leq 0}$. By Lemma 5.8, we know $A_n \in \mathcal{D}^b(\mathcal{C}^\heartsuit)_{\geq 0}$, $A'_n \in \mathcal{D}^b(\mathcal{C}^\heartsuit)_{\leq 0}$ and

$$\text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(A_n, A'_n) \simeq \text{Map}_{\mathcal{C}^\heartsuit}(\pi_0 A_n, \pi_0 A'_n) \rightarrow \text{Map}_{\mathcal{C}}(A_n, A'_n) \simeq \text{Map}_{\mathcal{C}^\heartsuit}(\pi_0 A_n, \pi_0 A'_n)$$

is equivalence.

- Let $(A_\bullet, X_{\bullet,\bullet}, B_\bullet, Y_{\bullet,\bullet}) \in \text{Gap}^\vee(\Phi)$. By Lemma 5.8, there is a cofiber sequence

$$\text{cofib}(\tau_{\geq 0} X_{0,i} \rightarrow \tau_{\geq 0} X_{0,i+1}) \rightarrow \tau_{\geq 0} \Phi(A_{i+1}) \rightarrow \pi_0 A_{i+1} / \text{coker}(\pi_0 X_{0,i} \rightarrow \pi_0 X_{0,i+1})$$

for each $i \geq 0$. Hence, by Lemma 5.8,

$$\text{Map}_{\mathcal{C}}(\tau_{\geq 0} \Phi(A_{i+1}), \pi_0 A_{i+1} / \text{coker}(\pi_0 X_{0,i} \rightarrow \pi_0 X_{0,i+1})) \simeq \text{Map}_{\mathcal{D}^b(\mathcal{C}^\heartsuit)}(\tau_{\geq 0} A_{i+1}, \pi_0 A_{i+1} / \text{coker}(\pi_0 X_{0,i} \rightarrow \pi_0 X_{0,i+1})).$$

Then, there is a morphism $\tau_{\geq 0} A_{i+1} \rightarrow \pi_0 A_{i+1} / \text{coker}(\pi_0 X_{0,i} \rightarrow \pi_0 X_{0,i+1})$ whose image under Φ is $\tau_{\geq 0} \Phi(A_{i+1}) \rightarrow \pi_0 A_{i+1} / \text{coker}(\pi_0 X_{0,i} \rightarrow \pi_0 X_{0,i+1})$. Thus,

$$\Phi(\text{fib}(\tau_{\geq 0} A_{i+1} \rightarrow \pi_0 A_{i+1} / \text{coker}(\pi_0 X_{0,i} \rightarrow \pi_0 X_{0,i+1}))) \simeq \text{cofib}(\tau_{\geq 0} X_{0,i} \rightarrow \tau_{\geq 0} X_{0,i+1}).$$

Similarly, there is a morphism $\text{fib}(\tau_{\geq 0} B_{i+1} \rightarrow \pi_0 B_{i+1} / \text{coker}(\pi_0 Y_{0,i} \rightarrow \pi_0 Y_{0,i+1}))$ such that

$$\Phi(\text{fib}(\tau_{\geq 0} B_{i+1} \rightarrow \pi_0 B_{i+1} / \text{coker}(\pi_0 Y_{0,i} \rightarrow \pi_0 Y_{0,i+1}))) \simeq \text{cofib}(\tau_{\geq 0} Y_{0,i} \rightarrow \tau_{\geq 0} Y_{0,i+1})$$

Define

$$\tau_{\geq 0}(A_\bullet, X_{\bullet,\bullet}, B_\bullet, Y_{\bullet,\bullet}) := \begin{cases} \text{fib}(\tau_{\geq 0} A_{i+1} \rightarrow \pi_0 A_{i+1} / \text{coker}(\pi_0 X_{0,i} \rightarrow \pi_0 X_{0,i+1})) \\ 0 \rightarrow \tau_{\geq 0} X_{0,1} \rightarrow \tau_{\geq 0} X_{0,2} \rightarrow \cdots \\ \text{fib}(\tau_{\geq 0} B_{i+1} \rightarrow \pi_0 B_{i+1} / \text{coker}(\pi_0 Y_{0,i} \rightarrow \pi_0 Y_{0,i+1})) \\ 0 \rightarrow \tau_{\geq 0} Y_{0,1} \rightarrow \tau_{\geq 0} Y_{0,2} \rightarrow \cdots \end{cases}$$

and $\tau_{\leq -1}(A_\bullet, X_{\bullet,\bullet}, B_\bullet, Y_{\bullet,\bullet}) := \text{cofib}(\tau_{\geq 0}(A_\bullet, X_{\bullet,\bullet}, B_\bullet, Y_{\bullet,\bullet}) \rightarrow (A_\bullet, X_{\bullet,\bullet}, B_\bullet, Y_{\bullet,\bullet}))$.

□

Define $\mathcal{F}_n : \mathcal{A}_n \rightarrow \mathcal{C}_n$:

- $\mathcal{C}_1 := \mathcal{C}, \mathcal{A}_1 := \mathcal{D}^b(\mathcal{C}^\vee), \mathcal{F}_1 := \Phi.$
- For $n \geq 2$, $\mathcal{C}_n := \text{Gap}^\vee(\mathcal{F}_{n-1}), \mathcal{A}_n := \text{Gap}^\vee(\mathcal{A}_{n-1})$ and $\mathcal{F}_n := \text{Gap}_{\mathcal{F}_{n-1}}^\vee.$

Define abelian categories $\mathcal{E}_n$:

- $\mathcal{E}_1 = \mathcal{C}^\vee.$
- $\mathcal{E}_n = \text{Seq}^\vee(\mathcal{E}_{n-1}).$

Theorem 5.10. $\Phi : \mathcal{D}^b(\mathcal{C}^\vee) \rightarrow \mathcal{C}$ is fillable and satisfies the dévissage condition. Hence, there are isomorphisms

$$K_n(\mathcal{D}^b(\mathcal{C}^\vee)) \xrightarrow{\cong} K_n(\mathcal{C})$$

for all $n \geq 0$.

Proof. By Corollary 5.5 and Proposition 5.9, we have $\mathcal{A}_n \simeq \mathcal{D}^b(\mathcal{E}_n)$, $\mathcal{C}_n$ admits a bounded t-structure with heart $\mathcal{E}_n$, and $\mathcal{F}_n$ is the canonical t-exact functor $\mathcal{D}^b(\mathcal{C}_n^\vee) \rightarrow \mathcal{C}_n$.

Then by Lemma 5.6, $\mathcal{F}_n : \mathcal{D}^b(\mathcal{C}^\vee) \rightarrow \mathcal{C}$ satisfies the dévissage condition. By Proposition 4.9, Φ is fillable and satisfies the dévissage condition.

By Corollary 4.12, there are isomorphisms

$$K_n(\mathcal{D}^b(\mathcal{C}^\vee)) \xrightarrow{\cong} K_n(\mathcal{C})$$

for all $n \geq 0$. □

Remark 5.11. Along with [Efi25], we have the following relationship:

$$\boxed{\text{Theorem 4.12}} \implies \boxed{\text{the theorem of heart}} \implies \boxed{\text{Quillen's dévissage theorem}}$$

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